MassEnviroScreen Documentation

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Introduction

This document presents a tool for identifying and prioritizing the most environmentally vulnerable or burdened communities in Massachusetts based on a cumulative burden score that incorporates exposure to pollution and climate risks and the presence of sensitive or vulnerable populations, as well as whether the community meets a criterion related to lower household income.

This tool - MassEnviroScreen - is a hybrid approach that utilizes an element of the current Massachusetts Environmental Justice Population definition, and a cumulative burden mapping approach. The cumulative burdens mapping approach is modeled on those used by California EPA's CalEnviroScreen tool and the Colorado EnviroScreen tool. The California and Colorado tools utilize a 'cumulative impact score' to describe the relative environmental burden of communities across the state and to prioritize those that are most burdened. California defines cumulative impacts as "the exposures, public health or environmental effects from the combined emissions and discharges, in a geographic area, including environmental pollution from all sources, whether single or multi-media, routinely, accidentally, or otherwise released." Impacts consider "sensitive populations and socio-economic factors, where applicable and to the extent data are available." The Colorado tool augments the CalEnviroScreen approach by adding climate risks, which is the approach followed here.

A 'cumulative burden score' is a numerical value that ranks every community (i.e., census block group) on a scale from 0 to 100. Higher values indicate greater cumulative burden. These values also represent percentile ranks, which means that a community's score indicates the percentage of scores in Massachusetts that are equal to or lower than a given score. For example, a census block group with a score of 75 (75th percentile) means that its cumulative burden score is equal to or higher than 75% of census block groups in the state. In this model, we follow California's example of using a score of 75 (the 75th percentile) as one of the thresholds for identifying the most impacted or 'Burdened Areas.' Census block groups with a minimum score of 75 experience cumulative burdens that are equal to or higher than 75% of block groups in the state. In other words, these block group represent the top 25% of cumulative burden scores in Massachusetts.

To define "Burdened Areas" in this tool, cumulative burden percentile scores of 75 or higher are combined with the existing Massachusetts EJ criterion for low income.

Burdened Areas are communities (i.e., census block groups) that meet one or more of the following criteria:

- cumulative burden percentile score (i.e, MassEnviroScore) of 75 or greater, OR

- annual median household income is 65 percent or less of the statewide annual median household income

MassEnviroScreen Scoring Methodology

The MassEnviroScreen cumulative burden score model is based on three components that represent Pollution and Climate Burden – Exposures, Environmental Effects, and Climate Risks – and two components that represent Population Characteristics – Sensitive Populations (e.g., in terms of health status and age) and Socioeconomic Factors.

Model characteristics

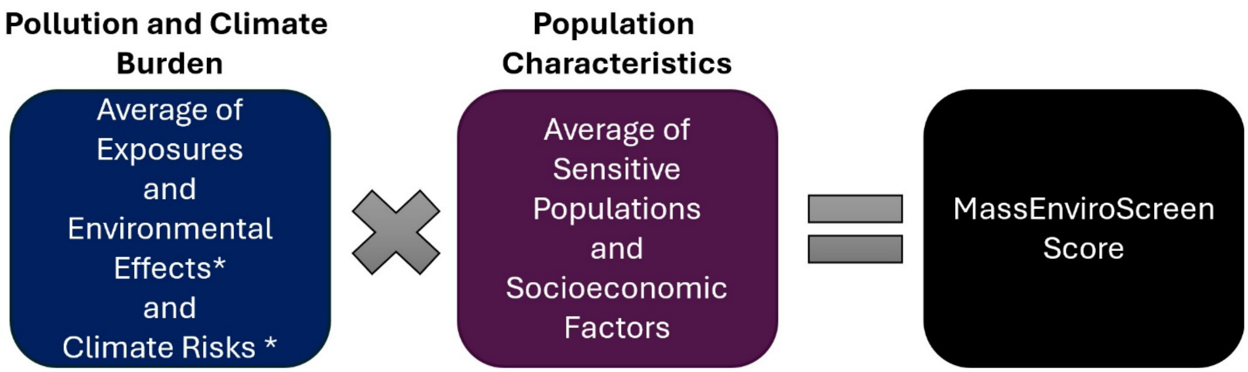
The model:

- Uses 30 statewide indicators to characterize Pollution and Climate Burden and Population Characteristics
- Uses percentiles to assign scores for each of the indicators in a given geographic area. The percentile represents a relative score for the indicators. **Please note: A higher percentile score does not necessarily mean that the indicator exceeds regulatory thresholds or poses direct human health risks at that score.**
- Uses a scoring system in which the percentiles are averaged for the set of indicators in each of the five components (Exposures, Environmental Effects, Climate Risks, Sensitive Populations, and Socioeconomic Factors).
- Combines the component scores to produce a MassEnviroScreen score for a given area relative to other areas in the state, using the formula below.

Formula for calculating MassEnviroScreen Score

After the components are scored within Pollution and Climate Burden and Population Characteristics, the scores are combined as follows to calculate the overall MassEnviroScreen Score:

Figure 1 – Formula for MassEnviroScreen cumulative burden score



* The Environmental Effects and Climate Risks scores were weighted half as much as the Exposures score.

Scores for the Pollution and Climate Burden and Population Characteristics categories are multiplied (rather than added, for example). This multiplication approach is based on research which shows that socioeconomic and health/sensitivity factors are "effect modifiers" that multiply the risks posed by pollutants and climate risks.[1] Various environmental health and emergency response organizations use a similar scoring system:

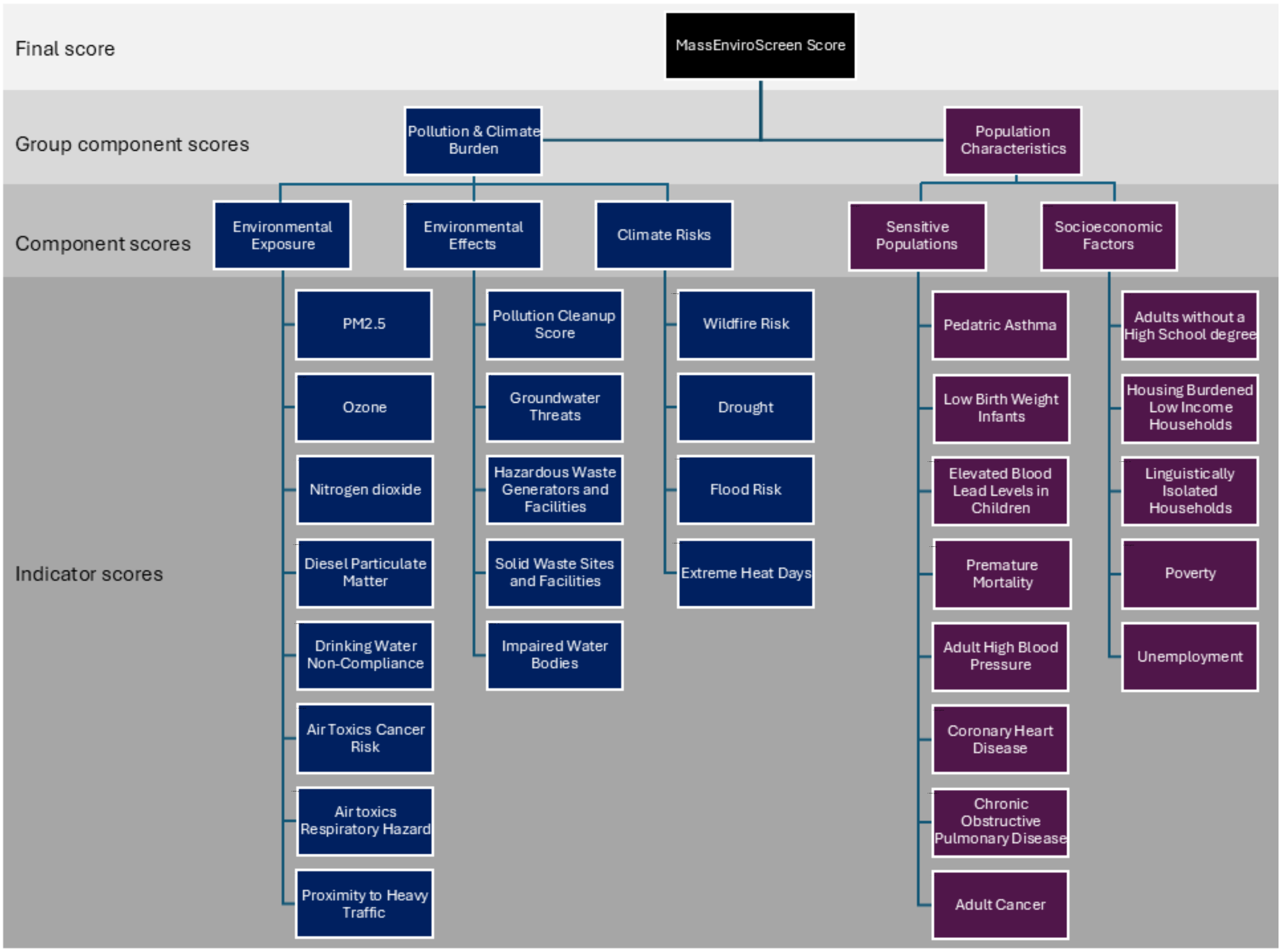
Risk = Threat x Vulnerability.[2]

Within the MassEnviroScreen score map, clicking on individual census block groups shows both the final MassEnviroScreen score as well as the Pollution and Climate Burden and the Population Characteristics SubScores. The Pollution and Climate Burden SubScore represents the percentile rank (0 - 100) of the block group for averaged Pollution and Climate Burden component indicators. Similarly, the Population Characteristics SubScore represents the percentile rank (0 - 100) of the block group for averaged sensitive populations and socioeconomic factors component indicators.

Model indicators

The MassEnviroScreen score model is computed from 30 statewide environmental, socioeconomic, and health indicators. These indicators are listed by indicator category in Figure 1 below. Details and data sources for model indicators are described later in this document.

Figure 2 - MassEnviroScreen Score Indicators



Example Census Block Group

Indicator Results, MassEnviroScreen Score, and designation as Burdened Area

One example census block group in Springfield was selected to illustrate how an overall MassEnviroScreen score is calculated and how a community is designated as a Burdened Area (BA).

Shown below are:

- A map of a census block group in the Six Corners neighborhood of Springfield, Massachusetts (highlighted in dark red). The block group GEOID is 250138019012.
- Tables for the indicators of Pollution and Climate Burden and Population Characteristics with raw and percentile scores for each of the indicators. Please note: These tables are illustrative and do not reflect how the raw values and percentiles will appear on the MassEnviroScreen tool.
- A table showing how a MassEnviroScreen score was calculated for this block group.

Figure 3 - Example Census Block Group

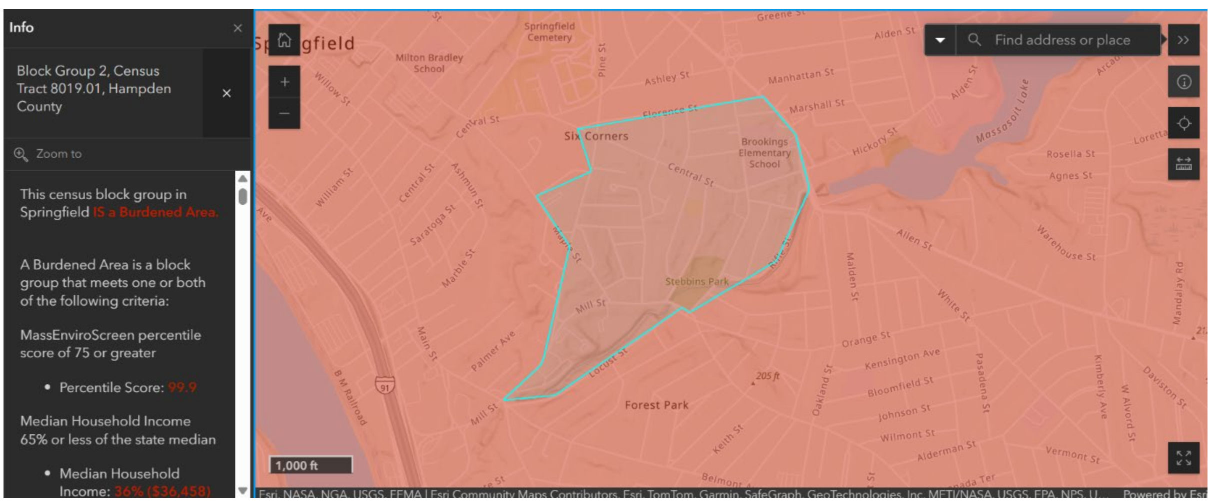


Figure 4 - Example Score for Pollution and Climate Burdens

Block Group 2, Census Tract 8019.01, Hampden County, City of Springfield, "Six Corners Neighborhood"

Exposure Indicators	Raw Value	Percentile
PM2.5	7.83 μg/m^3	99
Ozone (O_3)	58.39 ppb	80
Nitrogen Dioxide (NO_2)	13.06 ppb	84
Diesel Particulate Matter	0.27 μg/m^3	82
Drinking Water Non-Compliance (index)	6	77
Air Toxics Cancer Risk	28.5	96
Air Toxics Respiratory Hazard Index	0.3	80
Proximity to Heavy Traffic (impact index)	4,650,593	53
AVERAGE COMPONENT SCORE		**81.4**

Effects Indicators	Raw Value	Percentile
Pollution Cleanup Sites (weighted count)	66	98
Groundwater Threats (weighted count)	0	0
Hazardous Waste Generators and Facilities (weighted count)	0	0
Solid Waste Sites and Facilities (weighted count)	0	0
Impaired Water Bodies (count of pollutants)	2	65
AVERAGE COMPONENT SCORE		**32.5**

Climate Risks	Raw Value	Percentile
Drought (index)	1,988.7	23
Wildfire Risk (index)	6	60
Flood Risk	0%	0
Extreme Heat Days	460.5	97
AVERAGE COMPONENT SCORE		**45.1**

Figure 5 - Example Scores for Population Characteristics

Block Group 2, Census Tract 8019.01, Hampden County, City of Springfield, "Six Corners Neighborhood"

Sensitive Populations	Raw Value	Percentile
Pediatric Asthma (prevalence per 100 students K-8)	13.4%	76
Adult High Blood Pressure (prevalence)	38.8	97
Adult Cancer (prevalence)	3.9	4
Coronary Heart Disease (prevalence)	8.5	88
Low Birth Weight Infants (percent of live births)	3.8%	88
Chronic Obstructive Pulmonary Disease (prevalence)	10.4	97
Elevated Blood Lead Levels in Children (prevalence per 1,000 children 9 – 48 mos)	41.6	94
Premature Mortality (per 100,000 residents)	538.2	94
AVERAGE COMPONENT SCORE		**79.7**

Socioeconomic Factors	Raw Value	Percentile
Adults without a High School Degree	40.7%	98
Housing Burdened Low Income Households	35.7%	95
Linguistic Isolation	18.2%	89
Poverty	42%	97
Unemployment	15.4%	94
AVERAGE COMPONENT SCORE		**94.7**

Figure 6 - Example Calculation of MassEnviroScreen Score

Block Group 2, Census Tract 8019.01, Hampden County, City of Springfield, “Six Corners Neighborhood”

	Pollution& Climate Burden			**Population Characteristics**	
	Exposure	Environmental Effects	Climate Risks	Sensitive Populations	Socioeconomic Factors
Component Score (average of scaled indicators)	81.4	$(0.5 \times 32.5) = 16.3$	$(0.5 \times 45.1) = 22.6$	79.7	94.7
Average of Component Scores	$(81.4 + 16.3 + 22.6) \div (1 + 0.5 + 0.5) = \mathbf{60.15}$ Pollution & Climate Burden is calculated as the average of its three component scores, with the Environmental Effects and Climate Risks components half-weighted			$(79.7 + 94.7) \div 2 = \mathbf{87.2}$ Population Characteristics is calculated as the average of its two component scores	
Scaled Component Scores	$(60.15 \div 85.04) \times 10 = \mathbf{7.1}$ The Pollution & Climate Burden percentile is rescaled by the statewide maximum Pollution Burden scores			$(87.2 \div 94.6) \times 10 = \mathbf{9.2}$ The Population Characteristics percentile is scaled by the statewide maximum Population Characteristics scores	
MassEnviroScreen Score	$7.1 \times 9.2 = \mathbf{65.3}$ A score of **65.3** puts this census block group in the **99th percentile** or top 1% of all MassEnviroScreen scores statewide				

One of the thresholds for designation as a BA is a MassEnviroScreen cumulative burden score equal to or greater than the 75th percentile for the state. At the 99th percentile, this census block group is well above that threshold and therefore qualifies as a BA.

A census block group may qualify as a BA if it meets any of the following criteria:

- ☑ MassEnviroScreen cumulative impact percentile score of 75 or greater, OR
- ☑ annual median household income is 65 percent or less of the statewide annual median household income

In addition to the MassEnviroScreen cumulative burden score, this census block group meets the criterion for lower household income. Although a census block group need only exceed one of these thresholds to qualify as a BA, this example block group in Springfield exceeds both thresholds – 75th percentile MassEnviroScreen cumulative burden score and low median household income.

MassEnviroScreen Indicator Descriptions

For each indicator, this section provides the following details: Indicator name, definition, short description, which MassEnviroScreen component group it contributes to, the rationale (why it was selected as an indicator), the method (outlining assumptions and decisions that were made within data processing), years or time period of data source, the original geographic unit or spatial scale of the data, and the data source.

For more information about the original datasets, refer to the data source website/s included in each description.

Pollution and Climate Burden - Environmental Exposures

PM2.5

Definition: Average annual 24-hour average concentration of particulate matter that is less than or equal to 2.5 micrometers in diameter ($PM_{2.5}$) measured in micrograms per cubic meter (μg/m^3).

Standard: The federal and Massachusetts health-based standard for annual average PM2.5 is 9.0 μg/m^3 – all of Massachusetts meets this standard. The standard is set by USEPA (and adopted by MassDEP) to protect public health with an adequate margin of safety, including the health of sensitive populations.

Description: PM2.5 is a name for tiny particles (all particles less than 2.5 micrometers in diameter). These particles come from many different sources and have different chemical properties. Sources include vehicle tailpipes, smokestacks, dust from construction sites, fires, and chemical reactions in the atmosphere.

Indicator type: *Environmental Exposures*

Rationale: PM2.5 is one of six widespread air pollutants for which there are national air quality standards to limit their levels in the outdoor air. PM2.5 is an air pollutant that has been associated with the largest disease burden worldwide. PM2.5 is a group of particles from different sources and chemical components. Due to their small size, they have the capacity to be inhaled, travel to the lower respiratory tract, cross the lung-blood barrier, and be transported through the blood to any organ and system in the human body.[3] PM2.5 has been associated with multiple diseases and causes of death in adults and children. PM2.5 has the capacity to produce short and long-term health effects. In the short term, PM2.5 has been linked to pneumonia, cardiac arrhythmia, cardiac arrest, and mortality. In the long-term, PM2.5 exposure in children has been associated with asthma; PM2.5 exposure in pregnant women has been associated with adverse pregnancy outcomes; and PM2.5 exposure in adults has been associated with Alzheimer's disease, chronic kidney disease, chronic obstructive pulmonary disease, dementia, depression, ischemic heart disease, lung cancer, liver cancer, colorectal cancer, respiratory disease, stroke, type 2 diabetes, Parkinson's disease, and mortality, among others.[4] Some people are at disproportionately higher risk of experiencing health effects from long term exposure to PM2.5, including minorities, children, people with preexisting cardiovascular and respiratory disease, the overweight/obese, current/former smokers, and people with low socioeconomic status.[5]

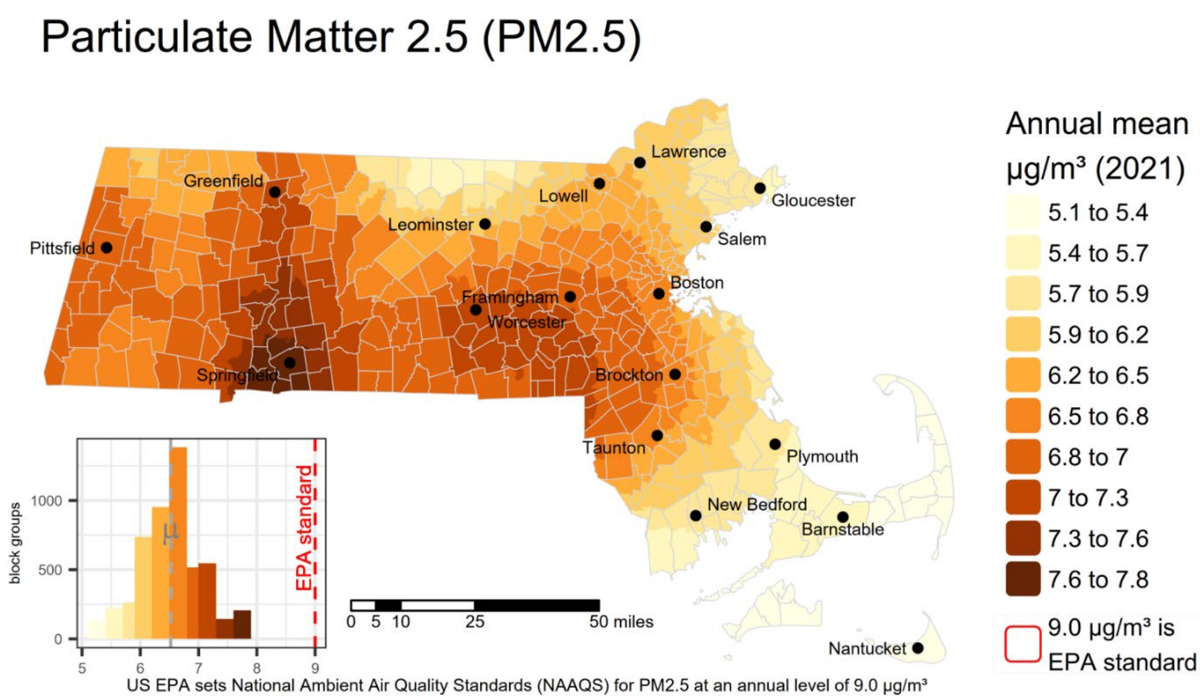
Method: To estimate PM2.5 concentrations in each census tract USEPA used PM2.5 air quality monitoring data from the State and Local Air Monitoring Stations (SLAMS) and numerical output from the Community Multiscale Air Quality (CMAQ) model into a Bayesian Space-time Downscaling Fusion Model (Downscaler). All census block groups received the same PM2.5 measure as the census tract in which they are contained.

Data years: 2021

Originating geographic scale of data: census tracts

Source: U.S. Environmental Protection Agency (EPA) Bayesian Space-time Downscaling Fusion Model (Downscaler) via EJScreen 2024. https://www.epa.gov/hesc/rsig-related-downloadable-data-files#output or https://screening-tools.com/epa-ejscreen

Figure 7 - Map of raw PM2.5 values



Ozone (O3)

Definition: Maximum 8-hour average model predictions of concentrations of ground-level ozone in parts per billion (ppb).

Standard: The federal and Massachusetts health-based standard for ozone 8-hour average is 70 ppb – all of Massachusetts meets this standard. The standard is set by USEPA (and adopted by MassDEP) to protect public health with an adequate margin of safety, including the health of sensitive populations.

Description: Ground-level ozone is formed from the reaction of oxygen-containing compounds with other air pollutants in the presence of sunlight. Ozone levels are typically at their highest in the afternoon and on hot days.

Indicator type: *Environmental Exposures*

Rationale: Ozone is one of six widespread air pollutants for which there are national air quality standards to limit their levels in the outdoor air. Ozone is an extremely reactive form of oxygen. Ozone has the capacity to directly harm the respiratory tissues when it is inhaled, producing oxidative stress and inflammatory reactions. Ozone exposure has been associated with several acute and chronic respiratory diseases, such as asthma exacerbation, asthma, pneumonia, and respiratory mortality.[6] Over time, ozone can also produce a more systemic inflammatory process, which may contribute to heart disease and mortality, adverse pregnancy outcomes, and neurological diseases. Some people are at higher risk of experiencing health effects from both short-term and long-term exposure to ozone, including children, the elderly, and people with asthma.[7] In addition to its adverse impacts on human health, ozone can also cause harm to public welfare and the economy because it harms plant tissue and interferes with vegetation and crop growth.[8]

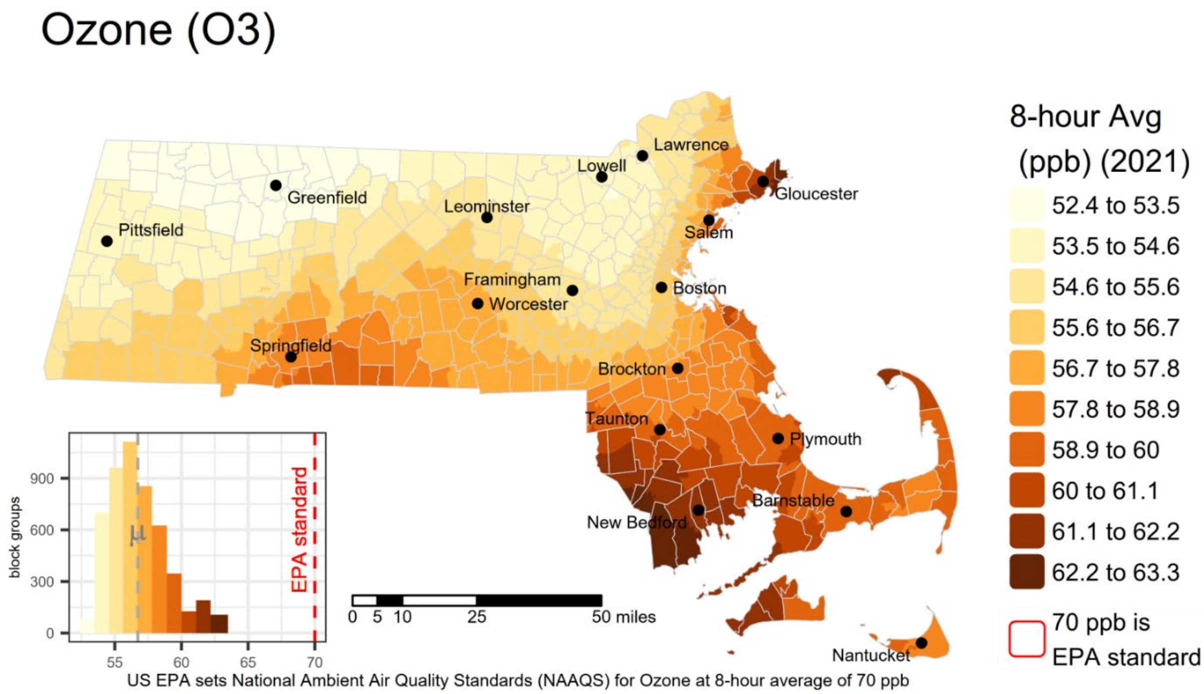
Method: To estimate ozone concentrations for each census tract USEPA used daily 8-hour maximum ozone concentrations measured in parts per billion (ppb) from the State and Local Air Monitoring Stations (SLAMS) and numerical output from the Community Multiscale Air Quality (CMAQ) model into the Bayesian Space-time Downscaling Fusion Model (Downscaler). All census block groups received the same ozone measure as the census tract in which they are contained.

Data years: 2021

Originating geographic scale of data: census tracts

Source: U.S. Environmental Protection Agency (EPA) Bayesian Space-time Downscaling Fusion Model (Downscaler) via EJScreen 2024. https://www.epa.gov/hesc/rsig-related-downloadable-data-files#output or https://screening-tools.com/epa-ejscreen

Figure 8 - Map of raw Ozone values



Nitrogen dioxide (NO2)

Definition: Average annual nitrogen dioxide (NO_2) levels expressed in parts per billion (ppb).

Standard: The federal and Massachusetts health-based standard for annual average NO_2 is 53 ppb – all of Massachusetts meets this standard. The standard is set by USEPA (and adopted by MassDEP) to protect public health with an adequate margin of safety, including the health of sensitive populations.

Description: Nitrogen dioxide, or NO_2, is a gaseous air pollutant composed of nitrogen and oxygen and is one of a group of related gases called nitrogen oxides, or NOx. NO_2 primarily gets into the air from the burning of fuel, such as gasoline, coal, oil, methane gas (natural gas), diesel, or wood. NO_2 forms from emissions from cars, trucks and buses, power plants, and off-road equipment and chemical reactions in the atmosphere.

Indicator type: *Environmental Exposures*

Rationale: Nitrogen dioxide is one of six widespread air pollutants for which there are national air quality standards to limit their levels in the outdoor air. Breathing air with NO_2 concentrations higher than the standard can irritate airways in the human respiratory system.[9] Such exposures over short periods can aggravate respiratory diseases, particularly asthma, leading to respiratory symptoms (such as coughing, wheezing or difficulty breathing), hospital admissions and visits to emergency rooms. Longer exposures to elevated concentrations of NO_2 may contribute to the development of asthma and potentially increase susceptibility to respiratory infections. People with asthma, as well as children and the elderly are generally at greater risk for the health effects of NO_2. NO_2 along with other NOx reacts with other chemicals in the air to form both particulate matter and ozone. NO_2 and other NOx interact with water, oxygen and other chemicals in the atmosphere to form acid rain. Acid rain harms sensitive ecosystems such as lakes and forests. NOx in the atmosphere contributes to nutrient pollution in coastal waters.[10]

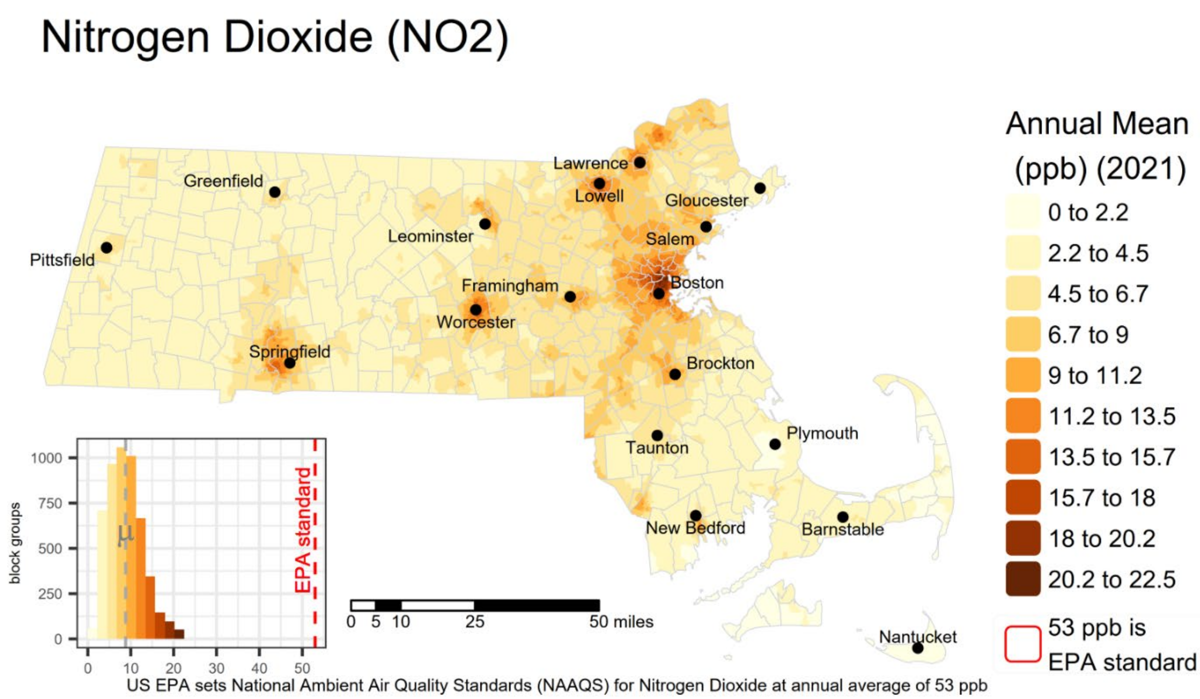
Method: Source data for the NO_2 indicator are developed by NASA's Health and Air Quality Applied Sciences Team (HAQAST). The NO_2 indicator is developed from a regression model, satellite data, and large-scale models. The resulting product is an approximately one square kilometer grid of NO_2 concentrations. These grids are then projected to the 2020 census block groups using a high-resolution NASA satellite product. The gridded data is then aggregated to census block group geographies using mean pixel values.

Data years: 2020

Originating geographic scale of data: 1km^2 pixels

Source: NASA's Health and Air Quality Applied Sciences Team (HAQAST) - Nitrogen Dioxide Surface-Level Annual Average Concentrations V1 via EJScreen 2024. https://disc.gsfc.nasa.gov/datasets/SFC_NITROGEN_DIOXIDE_CONC_1/summary?keywords=TROPOMI%20NO2 or https://screening-tools.com/epa-ejscreen

Figure 9 - Map of raw Nitrogen Dioxide values



Diesel Particulate Matter

Definition: Diesel particulate matter (PM) level in air measured in micrograms per cubic meter (μg/m^3).

Standard: Diesel PM does not have a federal or state health-based standard. USEPA does not include diesel PM in the air toxics cancer risk estimate. The Massachusetts average diesel PM level is 0.176, ranging from 0 to 0.5. The United States average diesel PM level is 0.2, ranging from 0 to 3.0.

Description: Diesel engines emit particles called diesel particulate matter or diesel PM, composed of hundreds of different chemicals. Diesel PM is different from non-diesel particulate matter evaluated as PM2.5. Diesel PM is highest near roadways, freeways, railyards, ports, and other industrial sites where diesel engines are used.

Indicator type: *Environmental Exposures*

Rationale: Short-term exposures to diesel PM can irritate the eyes, nose and throat; cause respiratory effects such as cough; and may exacerbate allergic responses.[11] Long-term exposures to diesel PM can lead to asthma, allergic responses, respiratory illnesses, cardiovascular and pulmonary diseases, and lung cancer. Some people may be at higher risk of experiencing health effects from exposure to diesel PM, including those with asthma, heart and lung diseases, as well as children and the elderly. This measure is included in addition to traffic proximity and the other air pollution indicators to reflect the unique exposure profile that diesel PM produces.

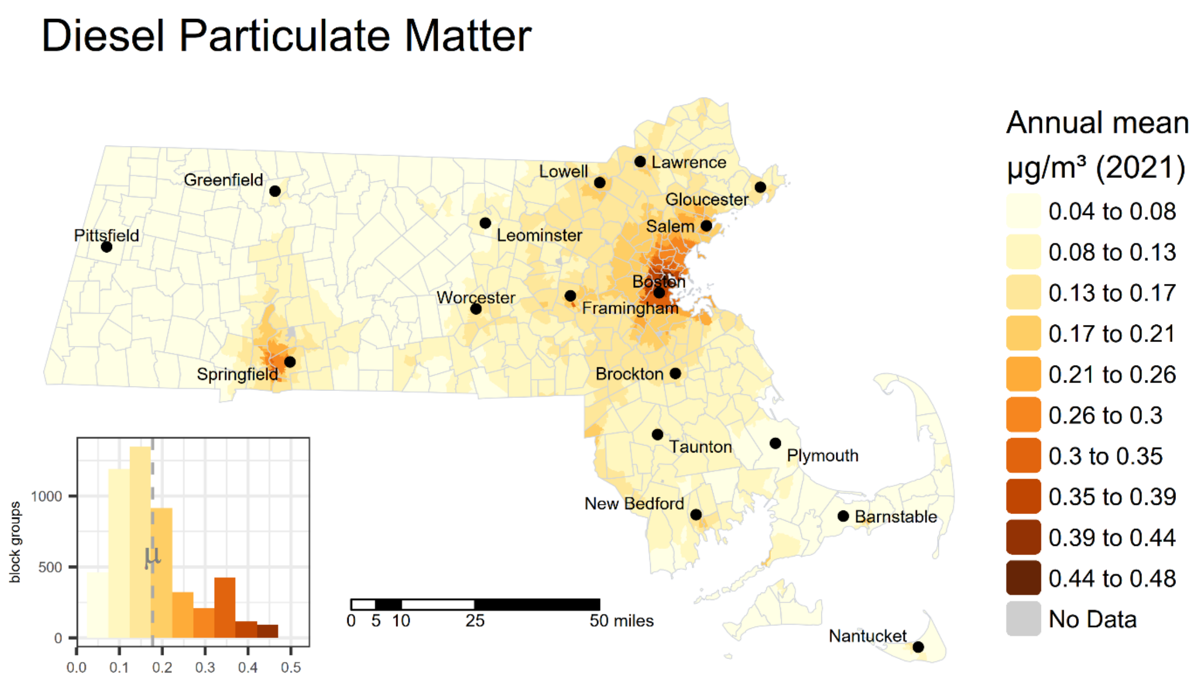
Method: The Diesel PM indicator is the estimated concentration of Diesel PM as provided by the 2020 AirToxScreen. The raw indicator is expressed in units of micrograms per cubic meter (μg/m^3), and reported at the census block group-level.

Data years: 2020

Originating geographic scale of data: census block group

Source: U.S. Environmental Protection Agency (EPA) Air Toxics Screening Assessment via EJScreen 2024. https://www.epa.gov/AirToxScreen or https://screening-tools.com/epa-ejscreen

Figure 10 - Map of raw Diesel Particulate Matter values



Drinking water Non-Compliance

Definition: Safe Drinking Water Act (SDWA) compliance performance score of a community water system (CWS) serving a census block group population.

Context: Community Public Water Systems are expected to comply with a diverse set of requirements within the National Primary Drinking Water Regulations and the corresponding Massachusetts Drinking Water Regulations. When they do not, violations are issued. This metric assigns a point value to each violation based on how quickly a consumer might see health impact(s) and the length of time the violation persists. Lower scores are better.

Description: The goal of this indicator is to highlight populations served by community water systems (CWSs) that have challenges complying with National Primary Drinking Water Regulations (NPDWRs). The higher the Drinking Water Non-Compliance Indicator score, the more compliance challenges faced by the CWS(s) serving that census block group population. The drinking water non-compliance indicator distills hundreds of different violation types into a single value that attempts to capture the federal Safe Drinking Water Act (SDWA) compliance performance of a CWS. The compliance data used by this indicator does not directly measure drinking water quality as this depends on violation type and other factors unique to each water system. The drinking water non-compliance indicator is only applicable to households that get their drinking water from a CWS, which are subject to National Primary Drinking Water Regulations (NPDWRs) under the SDWA. The indicator excludes households on private wells, for example, which are not regulated by the USEPA. The indicator reflects compliance performance for the last 5-years (2019 Q4- 2023 Q4). Scores are only given to water systems for violations that have not been returned to compliance.[12]

Indicator type: *Environmental Exposures*

Rationale: Public water systems must meet health-based federal standards for contaminants, including performing regular monitoring and reporting. The Public Water System Supervision or PWSS program is designed to protect public health by ensuring the safety of drinking water. Public water systems are regulated by USEPA, and primacy agencies (states, territories, and Tribes with EPA approval to implement the SDWA) and provide drinking water to 90% of the public. These public drinking water systems, which may be publicly- or privately-owned, serve at least 15 service connections or 25 persons for at least 60 days annually.

Method: Violations are divided into 3 categories with associated point values weighted by severity (based on the three SDWA public notice violation tiers):

1. Health Based Violations (immediate impacts), 10 points:
    a. Nitrate and/or Nitrite Maximum Contaminant Levels (MCLs), Acute Maximum Residual Disinfectant Level (MRDL), Revised Total Coliform Rule (R/TCR) Acute, Turbidity Treatment Technique (TT), Surface Water Treatment Rule Treatment Technique (SWTR TT).
2. Other Health Based Violations, 5 points:
    a. Any other health-based violation (violation type MCL, MRDL or TT), plus Total Coliform Rule TCR Monitoring/Reporting (MR) repeat violations (violation code 25 and 26) and Nitrate MR.
3. Non-Health Based Violations, 1 point:
    a. Any other Monitoring Violation (MON), Reporting Violation (RPT), Monitoring and Reporting (MR), or Other Violations categories.

For all 10- and 5-point violation types (see 1a. and 2a. above), the time (in years) a violation has been in non-compliance is added to the violation score. 1-point is added for every year of non-compliance, up to 5-years (5-points).

The violation data that makes up the drinking water non-compliance indicator was taken from EPA's internal Enforcement Targeting Tool (ETT) which draws upon the SDWA system of record known as the Safe Drinking Water Information System, or SDWIS. SDWIS is updated on a quarterly basis and the violation data used to create the Drinking Water Non-Compliance Indicator score is from 2023 Q4.

The 5-year service area score of all violation types is first appended into utility service area boundaries:

$$Service\ Area\ Score = \sum(Violation\ severity + Years\ in\ non-compliance_{health-based})$$

The indicator score is then calculated at the block level and weighted by block population as a percent of its parent block groups' population (see equation below). If any part of the service area boundary is within a block, that block's score gets associated with that service area boundary(ies). If two or more service areas intersect the same block the scores are averaged. This method takes the most health protective approach: theoretically, if only one person is served by a public water system in that block, the score is applied to everyone living in that census block.

$$DWInScore_{blkGrp} = \sum \frac{(DWInScore_{blk})}{n} \times \%\ of\ block\ group\ population$$

The block group score is the sum of the scores calculated at the associated census block level (see equation below). If only one system serves a block group, and the system encompasses all associated block group blocks, the block group score will be identical to the service area score for that system.

$$DWInScore_{blkGrp} = \sum (DWInScore_{blk})$$

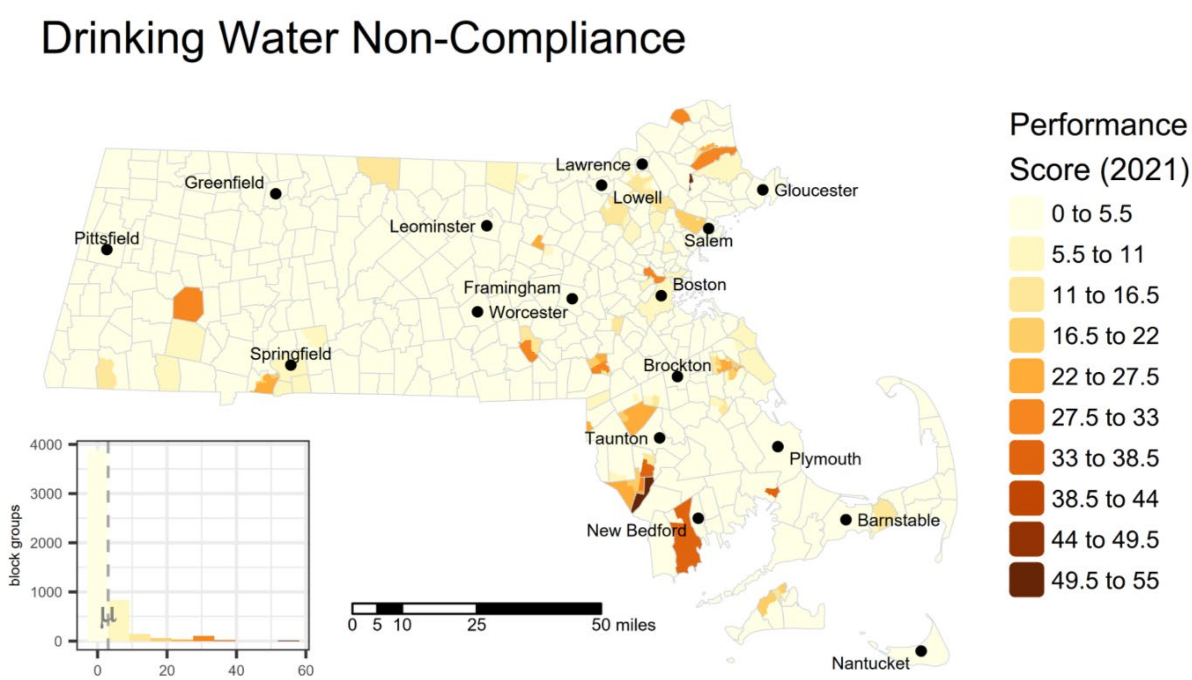
If no service areas intersect any part of a block group, that block group gets a null score. These areas are typically supplied by private well water.

Data years: 2019 - 2023

Originating geographic scale of data: census block groups

Source: EJScreen 2024 at https://screening-tools.com/epa-ejscreen

Figure 11 - Map of Drinking Water non-Compliance raw values



Air Toxics Cancer Risk

Definition: Risk of developing cancer due to inhalation exposure to a group of air toxics over a lifetime of 70 years estimated as cancer cases per million people.

Standard: Air toxics do not have pre-defined federal standards that clearly represent acceptable or unacceptable thresholds of additional lifetime cancer risk. The Massachusetts average lifetime cancer risk from air toxics as a group is 23 per million, ranging from 13 to 152 per million. The United States average lifetime cancer risk is 28 per million, ranging from 6 to 3,272 per million.

Description: USEPA has classified 188 pollutants as Hazardous Air Pollutants (HAPs) as defined in the Clean Air Act. Toxic, or hazardous, air pollutants include gases, such as hydrogen chloride, benzene and toluene; metals such as cadmium, mercury and chromium; and physical agents such as asbestos. Major sources of toxic air pollutants outdoors include emissions from coal-fired power plants, industries, and refineries, as well as from cars, trucks and buses.

Indicator type: *Environmental Exposures*

Rationale: Hazardous air pollutants, also known as toxic air pollutants or air toxics, are pollutants that are known or suspected to cause cancer or other serious health effects. The health effects from air toxics depend on the specific pollutant, the exposure levels, and the duration of exposure. People exposed to toxic air pollutants at sufficient concentrations and durations may have an increased chance of getting cancer or experiencing other serious health effects. Some people are at a higher risk of experiencing health effects from both short-term and long-term exposure to air toxics, including children, the elderly, and people already experiencing health challenges.

Method: AirToxScreen was developed by USEPA to provide a “snapshot” of outdoor air quality as it relates to hazardous air pollutants for each state in the continental US. AirToxScreen provides an estimate of the concentration of air toxics in outdoor air and calculates the associated cancer risk and non-cancer hazard for each individual air toxic and air toxics as a group for each census block group. The cancer risk and non-cancer hazard estimates assume that exposures are chronic (assumed to be a 70-year lifetime) and the concentrations estimated for the “snapshot” year remain the same during the lifetime.

AirToxScreen uses emissions data compiled for a single year as inputs to air quality models. The models use these source data along with meteorological data for the same year to estimate ambient air concentrations of specific air toxics. USEPA then combines these modeled concentrations with census data and other information to calculate exposure concentrations of the air toxics. AirToxScreen then estimates cancer risks and potential noncancer health effects associated with chronic inhalation exposure to the toxics. Detailed description of the AirToxicScreen methodology is available at https://www.epa.gov/AirToxScreen/2020-airtoxscreen-assessment-results .

USEPA's AirToxScreen follows several basic steps to produce the final assessment:

- Compile National Emissions Inventory;
- Estimate ambient concentrations of air toxics across the US;
- Estimate population exposures; and
- Characterize potential public health risks from inhalation exposure.

The National Emissions Inventory (NEI) is a detailed, nationwide inventory of air toxics emissions. The NEI includes emissions from point, nonpoint and mobile sources, as well as emissions from biogenic sources and fires. These source data form the USEPA's air emissions modeling platform and are the foundation of AirToxScreen's air quality modeling. NEI emissions and other necessary data (e.g., meteorological data), are used as inputs to two air quality models used to estimate ambient air concentrations of air toxics: the American Meteorological Society/Environmental Protection Agency Regulatory Model (AERMOD) atmospheric dispersion model and the Community Multiscale Air Quality (CMAQ) photochemical model. AERMOD is used for all AirToxScreen air toxics modeled, and CMAQ is used for a list of 52 air toxics that are incorporated into CMAQ multipollutant version 5.4.

AirToxScreen Air Toxics Cancer Risk data was processed using the following steps:

- Census block-level cancer risk by region and source group values were downloaded.
- To compute cancer risks resulting from exposure to multiple air toxics, chronic cancer risk for all air toxic source groups were summed for each census block, following AirToxScreen 2020 Documentation (p113 Sec 6.2.2. "Multiple-pollutant Risks"):

  $Risk_{tot} = Risk_1 + Risk_2 + ... + Risk_i$

  Where:

  $Risk_{tot}$ = total cumulative individual lifetime cancer risk, across *i* substances
- To aggregate cancer risk values at the census block group level, a population-weighted risk average of census blocks within each block group was calculated following AirToxScreen 2020 Documentation (p116 Sec 6.4.1. "Aggregation of Block-level Results to Larger Spatial Units"):

$$Conc_{blockgroup_k} = \frac{\sum Conc_{block_i} \times Pop_{block_i}}{Pop_{blockgroup_k}}$$

Where:

$Conc_{blockgroup_k}$= population-weighted concentration for block group *k*

$Conc_{block_i}$= ambient concentration in block *i* (contained within block group *k*)

Pop_{block_i} = population in block *i* (contained within block group *k*)

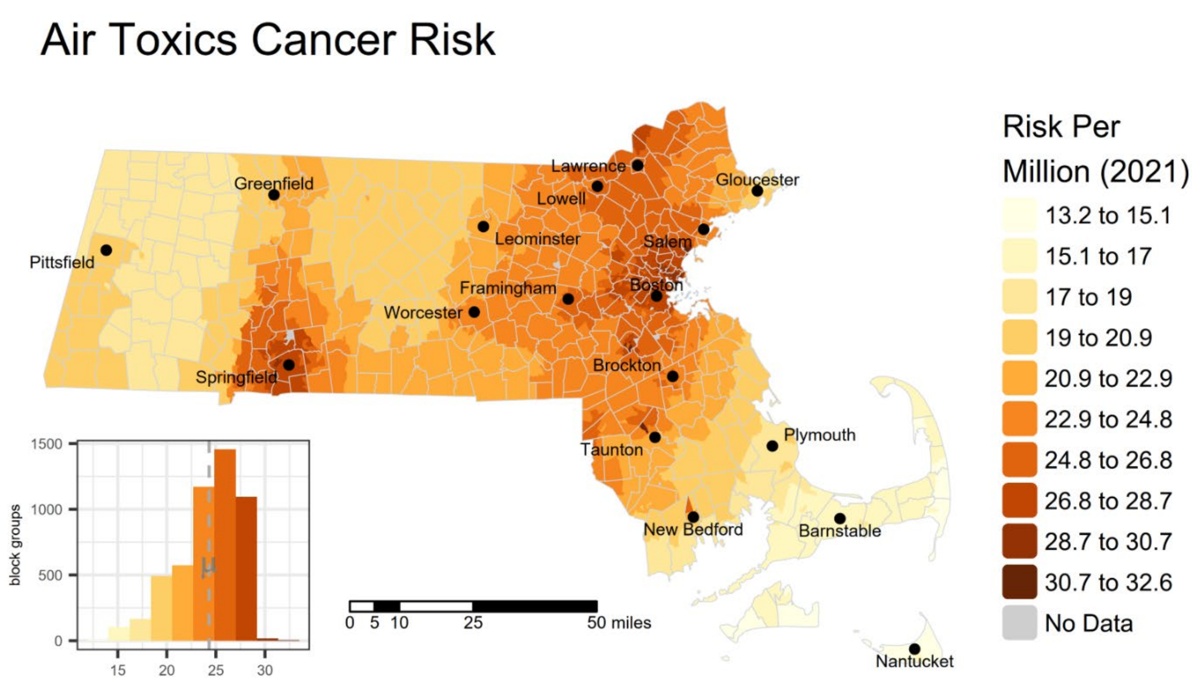
$Pop_{blockgroup_k}$= population in block group *k*

Data years: 2020

Originating geographic scale of data: census blocks

Source: US Environmental Protection Agency 2020 AirToxScreen: Assessment Results. https://www.epa.gov/AirToxScreen/2020-airtoxscreen-assessment-results

Figure 12 - Map of Air Toxics Cancer Risk raw values



Air Toxics Respiratory Hazard Index

Definition: Air Toxics Respiratory Hazard Index is an estimate of the potential for respiratory effects other than cancer from exposure to a group of air toxic pollutants. It is the sum of the respiratory effects for each air toxic estimated by the ratio of the exposure concentration to the health-based reference concentration for the specific air toxic.

Standard: Air toxics do not have pre-defined federal standards that clearly represent acceptable or unacceptable thresholds for the respiratory hazard index from a group of air toxics or for individual air toxics. However, a hazard index of one (1) or lower means air toxics are unlikely to cause adverse non-cancer respiratory health effects over a lifetime of exposure. The Massachusetts average air toxics respiratory hazard index is 0.26, ranging from 0.10 to 0.69. The US average air toxics respiratory hazard index from air toxics is 0.30, ranging from 0.05 to 3.91.

Description: USEPA has classified 188 pollutants as Hazardous Air Pollutants (HAPs) defined in the Clean Air Act. Toxic, or hazardous, air pollutants include gases such as hydrogen chloride, benzene and toluene, metals such cadmium, mercury and chromium, and physical agents such as asbestos. Major sources of toxic air pollutants outdoors include emissions from coal-fired power plants, industries, and refineries, as well as from cars, trucks and buses.

Indicator type: *Environmental Exposures*

Rationale: Hazardous air pollutants, also known as toxic air pollutants or air toxics, are pollutants that are known or suspected to cause cancer or other serious health effects. The health effects from air toxics depend on the specific pollutant, the exposure levels, and the duration of exposure. People exposed to toxic air pollutants at sufficient concentrations and durations may have an increased chance of getting cancer or experiencing other serious health effects. Some people are at a higher risk of experiencing health effects from both short-term and long-term exposure to air toxics, including children, the elderly, and people already experiencing health effects.

The Respiratory Hazard Index represents the sum of hazard quotients for the air toxics that target the respiratory system with health effects other than cancer. Past assessment results suggest that the respiratory endpoint (the effect of air toxics on the lungs and the rest of the respiratory system) usually drives non-cancer health effects. A hazard quotient is the ratio of the inhaled exposure concentration for each air toxic to its health based reference concentration (RfC) for inhalation. The hazard quotients are summed across all air toxics that have effects on the respiratory system to calculate the Respiratory Hazard Index.

A hazard index of 1 or lower means air toxics are unlikely to cause adverse non-cancer health effects over a lifetime of exposure. A hazard index value greater than 1 does not mean that an individual will experience health effects. It indicates an increased potential for health effects over a lifetime of exposure.

Method: AirToxScreen was developed by USEPA to provide a “snapshot” of outdoor air quality as it relates to hazardous air pollutants for each state in the continental US. AirToxScreen provides an estimate of the concentration of air toxics in outdoor air and calculates the associated cancer risk and non-cancer hazard for each individual air toxic and air toxics as a group for each census block group. The cancer risk and non-cancer hazard estimates assume that exposures are chronic (assumed to be a 70-year lifetime) and the concentrations estimated for the “snapshot” year remain the same during the lifetime.

AirToxScreen uses emissions data compiled for a single year as inputs to air quality models. The models use these source data along with meteorological data for the same year to estimate ambient air concentrations of specific air toxics. USEPA then combines these modeled concentrations with census data and other information to calculate exposure concentrations of the air toxics. AirToxScreen also estimates cancer risks and potential non-cancer health effects associated with chronic inhalation exposure to the toxics. Detailed description of the AirToxicScreen methodology is available at https://www.epa.gov/AirToxScreen/2020-airtoxscreen-assessment-results

USEPA’s AirToxScreen follows several basic steps to produce the final assessment:

- Compile National Emissions Inventory;
- Estimate ambient concentrations of air toxics across the US;
- Estimate population exposures; and
- Characterize potential public health risks from inhalation exposure.

The National Emissions Inventory (NEI) is a detailed, nationwide inventory of air toxics emissions. The NEI includes emissions from point, nonpoint and mobile sources, as well as emissions from biogenic sources and fires. These source data form the USEPA’s air emissions modeling platform and are the foundation of AirToxScreen’s air quality modeling. NEI emissions and other necessary data (e.g., meteorological data), are used as inputs to two air quality models used to estimate ambient air concentrations of air toxics: the American Meteorological Society/Environmental Protection Agency Regulatory Model (AERMOD) atmospheric dispersion model and the Community Multiscale Air Quality (CMAQ) photochemical model. AERMOD is used for all AirToxScreen air toxics modeled, and CMAQ is used for a list of 52 air toxics that are incorporated into CMAQ multipollutant version 5.4.

AirToxScreen Air Toxics Respiratory Hazard Index data was processed using the following steps:

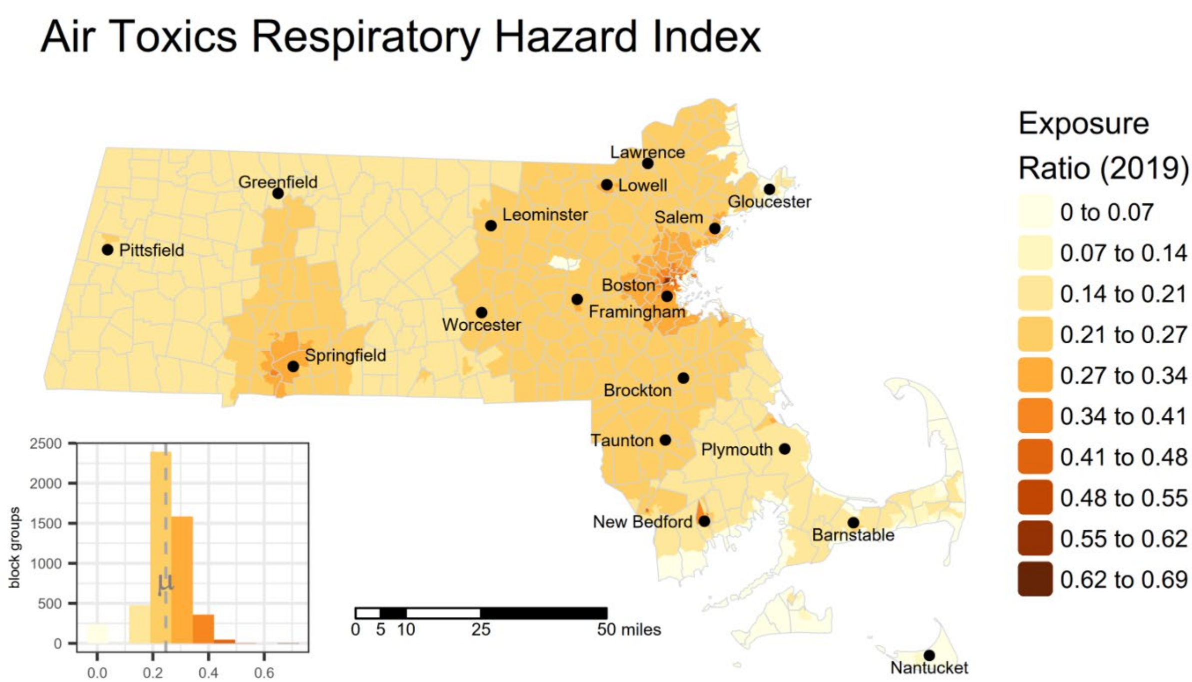
- Tract-level respiratory hazard index by region and source group values were downloaded.
- All census block groups received the same cumulative Respiratory Hazard Index measure as the census tract in which they are contained.

Data years: 2019

Originating geographic scale of data: census tracts

Source: US Environmental Protection Agency 2019 AirToxScreen: Assessment Results. https://www.epa.gov/AirToxScreen/2019-airtoxscreen-assessment-results

Figure 13 - Map of Air Toxics Respiratory Hazard Index raw values



Proximity to Heavy Traffic

Definition: Heavy traffic proximity impact index

Description: Heavy traffic proximity impact index is calculated as the sum of annual average daily traffic (AADT) at major roads within 10 kilometers of a census block centroid, divided by distance (major roads being interstates, principal arterials/other expressways, and minor arterials in urban areas).

Indicator type: *Environmental Exposures*

Rationale: Traffic is a significant source of air pollution, particularly in urban areas, where more than 50% of particulate emissions come from traffic. Exhaust from vehicles contains a number of toxic chemicals, including nitrogen oxides, carbon monoxide, and benzene.[13] Traffic exhaust also plays a role in the formation of ozone. Health effects of concern from these pollutants include heart and lung disease, cancer, and increased mortality.

Method: The traffic proximity indicator is based on AADT count divided by distance in meters from each census block centroid. The proximity score is based on the traffic within a search radius of 10 km. The closest traffic is given more weight, and the distant traffic is given less weight, through inverse distance weighting. The traffic proximity indicator values for each block centroid are averaged to the census block group in which they are contained.

Highway segments are from the HPMS lines and AADT counts are from the 2020 HPMS release, Federal Highway Administration (FWHA), U.S. Department of Transportation (DOT). Proximity scores were calculated by adding a 10km buffer to all non-zero population block groups; performing a one-to-many spatial join to identify all block group-segment pairs; assigning inverse distance scores for all identified 2020 Census blocks within each targeted block group with AADT multiplier (max score of 10); multiplying the scores by AADT and block population weights; and then aggregating the block scores to create a final block group score.

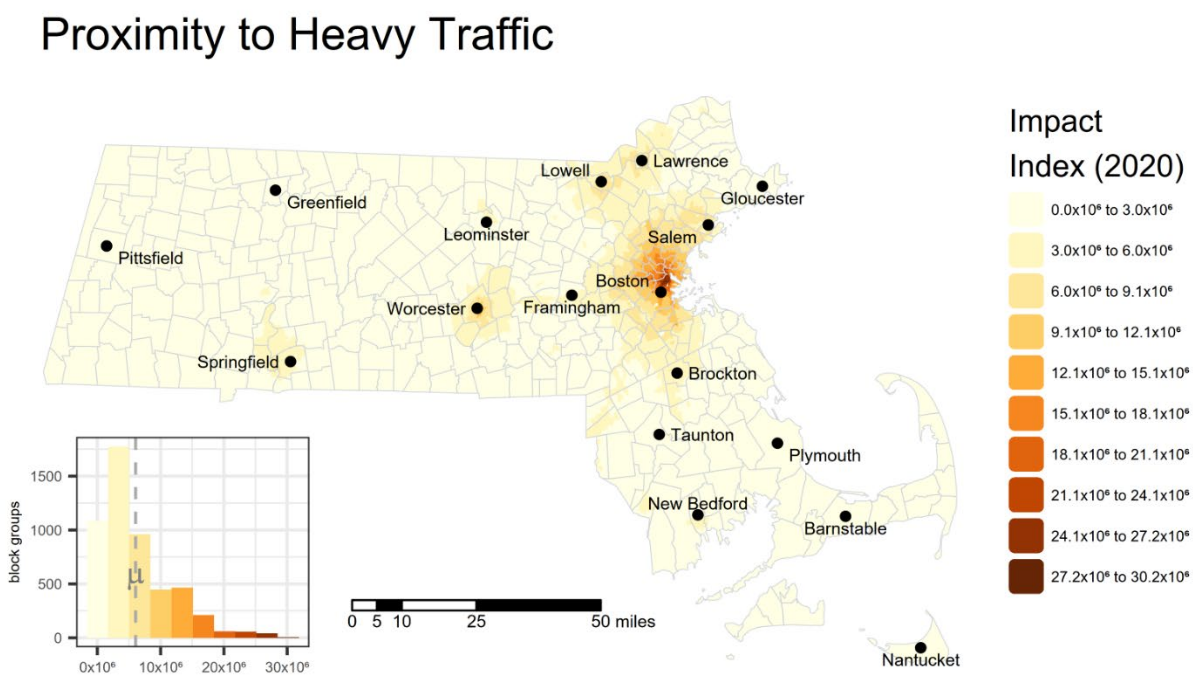
Data years: 2020

Originating geographic scale of data: census block groups

Indicator type: *Environmental Exposures*

Source: EJScreen 2024 at https://screening-tools.com/epa-ejscreen

Figure 14 - Map of Proximity to Heavy Traffic impact index raw values



Pollution and Climate Burden - Environmental Effects

Pollution Cleanup Sites

Definition: Weighted count of environmental cleanup sites requiring federal or state oversight for cleanup due to contamination.

Context: On average, there are about 1,000 new releases of oil or hazardous materials every year which need to be remediated. Types of releases vary from small oil spills to discovering historic contamination. Most releases are cleaned up within one year and over ninety percent of releases are cleaned up within six years.

Description: Sites undergoing cleanup actions by governmental authorities or by property owners have suffered environmental degradation due to the presence of hazardous substances. Of primary concern is the potential for people to come into contact with these substances. Some, such as “brownfield” sites, are also underutilized due to cleanup costs or concerns about liability. Since the nature and the magnitude of the threat and burden posed by hazardous substances vary among the different types of sites as well as the site status, the indicator takes both into account. Weights were also adjusted based on proximity to populated census blocks.

Indicator type: *Environmental Effects*

Rationale: Contaminated sites can pose a variety of risks to nearby residents. Hazardous substances can move off-site and impact surrounding communities through volatilization, groundwater plume migration, or windblown dust. Studies have found levels of organochlorine pesticides in blood[14] and toxic metals in house dust,[15] as well as PFAS that were correlated with residents’ proximity to contaminated sites.[16]

Method: Sites were scored on a weighted scale of 0 to 12 in consideration of both the site risk and status. Higher weights were applied to Superfund and Chapter 21E sites where there was known groundwater contamination or an imminent hazard. Similarly, higher weights were applied to sites that are undergoing active remediation and oversight, relative to those in which no further action was required. The weights for all sites were adjusted based on the distance they fell from populated census blocks. Sites further than 1000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for sites less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given census block group. Each census block group was scored based on the sum of the adjusted weights. Summed census block group scores were sorted and assigned percentiles based on their position in the distribution. The table below shows the weights for each site type.

Data years: National Priorities List and Superfund Alternative Approach Sites 2024; EPA ACRES brownfield sites 2024; MA DEP 21E sites 2025; MA DEP AUL sites 2025

Originating geographic scale of data: point locations

Weighting Matrix for Cleanup Sites

		Status		
		Low • Certified • Completed • No Further Action	Medium • Inactive – Needs Eval. • Certified Operation & Maintenance • Land Use Restrictions	High • Active • Inactive – Action Required
Risk	Low • No Significant Risk	**0**	**4**	**6**
	Medium • Not a substantial hazard but not "No Significant Risk" • Suspected Contamination	**1**	**7**	**9**
	High • Superfund • Groundwater contamination • Imminent hazard	**2**	**10**	**12**

Active: Identifies that an investigation and/or remediation is currently in progress and that DEP is actively involved, either in a lead or support capacity.

Certified Operation and Maintenance (O&M): Identifies sites that have certified cleanups in place but require ongoing O&M activities.

Certified: Identifies completed sites with previously confirmed releases that are subsequently certified by DEP as having been remediated satisfactorily under DEP oversight.

Inactive – Action Required: Identifies non-active sites where DEP has determined that a removal or remedial action or further extensive investigation is required.

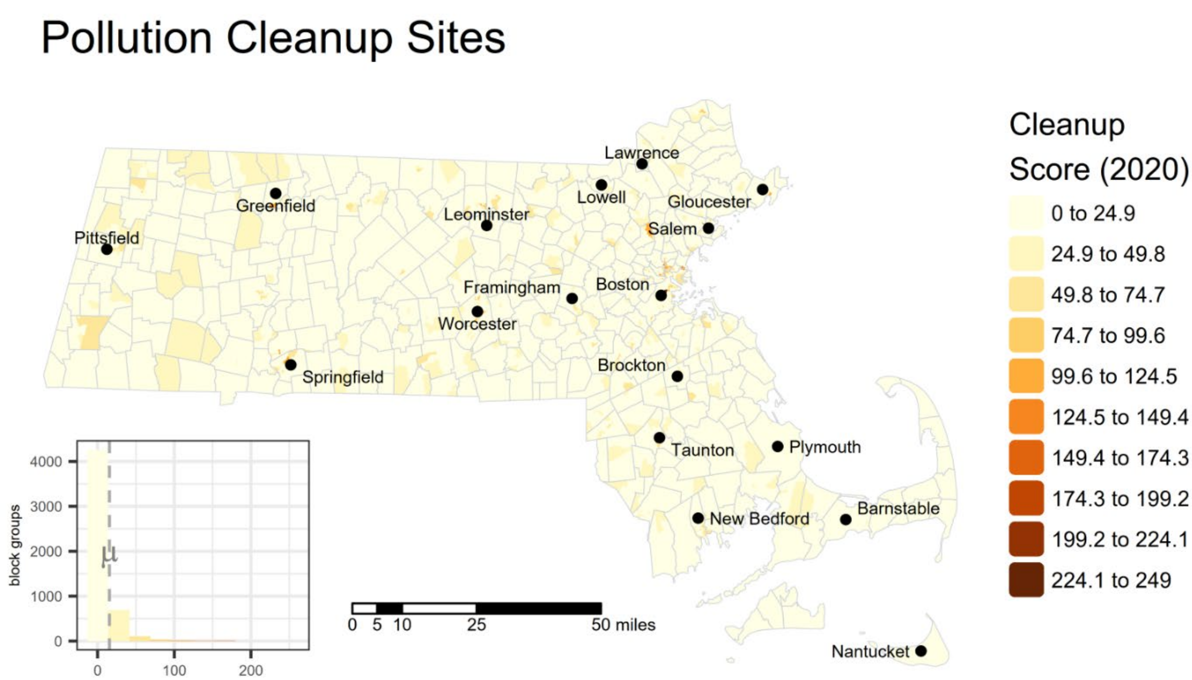
Inactive - Needs Evaluation: Identifies inactive sites where DEP has determined an evaluation is required.

No Further Action: Identifies completed sites where DEP determined that the property does not pose a problem to public health or the environment.

Superfund: Identifies sites where the US EPA proposed, listed, or delisted a site on the National Priorities List (NPL).

Sources: Superfund National Priorities List polygons from US EPA Office of Land and Emergency Management (OLEM) at https://edg.epa.gov/data/PUBLIC/OLEM/OLEM-OSRTI/NPL_Boundaries.zip; Brownfields sites from US EPA Assessment, Cleanup and Redevelopment Exchange System (ACRES) at https://catalog.data.gov/dataset/acres-brownfields-properties; MA DEP 21E sites from MassGIS at https://www.mass.gov/info-details/massgis-data-massdep-tier-classified-oil-andor-hazardous-material-sites-mgl-c-21e; MA DEP AUL sites from MassGIS at https://www.mass.gov/info-details/massgis-data-massdep-oil-andor-hazardous-material-sites-with-activity-and-use-limitations-aul;

Figure 15 - Map of Pollution Cleanup Sites raw values



Groundwater Threats

Definition: Weighted count of groundwater threats.

Description: Many activities can pose threats to groundwater quality. These include the storage and disposal of hazardous materials on land and in underground storage tanks at various types of commercial, industrial, and military sites. Sites included here include Underground Storage Tanks (USTs) and to a lesser degree facilities with groundwater discharge permits.

Indicator type: *Environmental Effects*

Rationale: The occurrence of storage tanks, leaking or not, provides a good indication of potential concentrated sources of some of the more prevalent compounds in groundwater. For example, the detection frequency of volatile organic compounds (VOCs) found in gasoline is associated with the number of UST sites within one kilometer of a well.[17] The occurrence of chlorinated solvents in groundwater is also associated with the presence of cleanup sites.[18] People who live near shallow groundwater plumes containing VOCs may also be exposed via the intrusion of vapors from soil into indoor air.[19]

Method: The weights for all sites were adjusted based on their distance from populated census blocks. Sites further than 1000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for sites less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given block group. Sites outside of a census block group, but less than 1000m from one of that block group's populated blocks were similarly adjusted based on the distance to the nearest block from that block group. Each census block group was scored based on the sum of the adjusted weights for sites it contains or is near.

Data years: USEPA UST Finder 2018-2021; MA DEP Groundwater Discharge Permits 2025

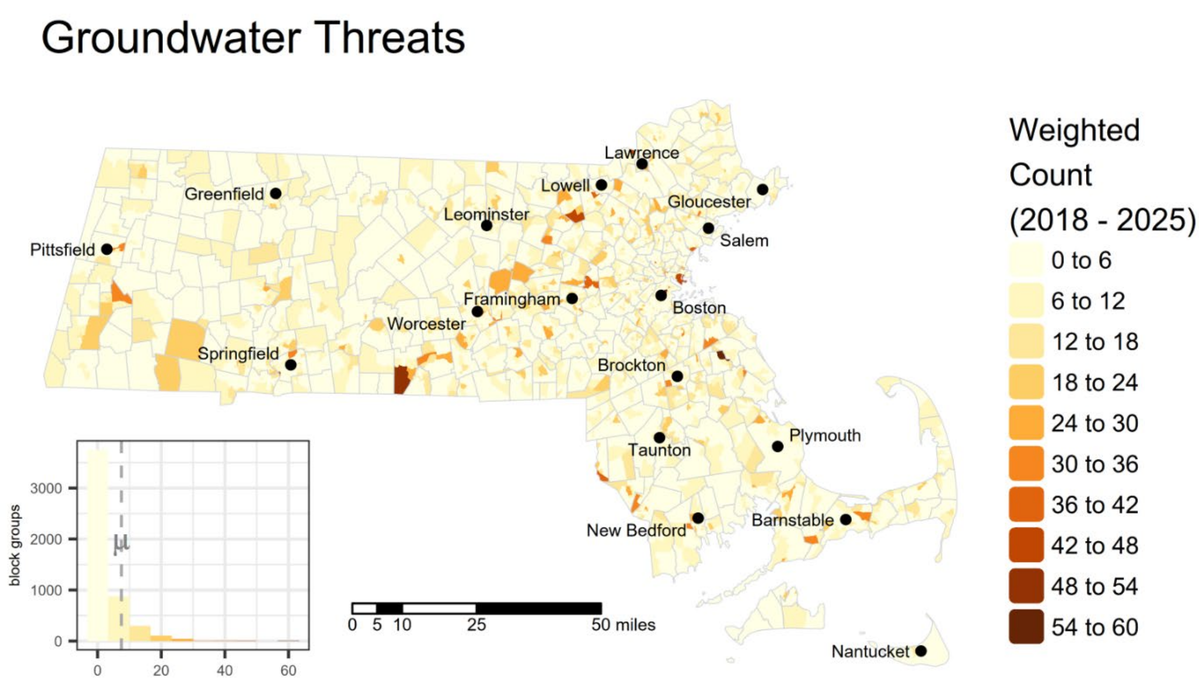
Originating geographic scale of data: point locations

Sources: US EPA's UST Finder data is a national composite of leaking underground storage tanks, underground storage tank facilities, and underground storage tanks as of 2018-2021. Data downloaded via ArcGIS Pro at https://epa.maps.arcgis.com/home/item.html?id=5a3ae0ed53564b6fa519f08e30e79e93; MA DEP Groundwater Discharge Permits from MassGIS at https://www.mass.gov/info-details/massgis-data-massdep-groundwater-discharge-permits

Weighting Matrix for Groundwater Threats

Site Type	Status/Type	Weight
Groundwater Discharge Permit Facilities	Industrial, Sanitary Discharge	5
	Car Wash, Laundromat, Reclaimed Water, Other	2
UST Sites	Open	3
	No Further Action	3
	Unknown	3

Figure 16 - Map of Groundwater Threats raw values



Hazardous Waste Generators and Facilities

Definition: Weighted count of hazardous waste facilities, and hazardous waste generators within each census block group.

Description: Most hazardous waste must be transported from hazardous waste generators to permitted recycling, treatment, storage, or disposal facilities (TSDF) by registered hazardous waste transporters. Shipments are accompanied by a hazardous waste manifest. There are widespread concerns for both human health and the environment from sites that serve to process or dispose of hazardous waste. Many newer facilities are designed to prevent the contamination of air, water, and soil with hazardous materials, but even newer facilities may negatively affect perceptions of surrounding areas in ways that have economic, social and health impacts. The Massachusetts Department of Environmental Protection (MassDEP) Bureau of Air and Waste maintains data on permitted facilities that are involved in the treatment, storage, or disposal of hazardous waste as well as information on hazardous waste generators. *Note that the presence of a hazardous waste facility does not mean that there has been any harmful exposure to hazardous waste in the surrounding neighborhoods. The potential for exposure, where it may exist, varies widely across facilities.*

Indicator type: *Environmental Effects*

Rationale: Hazardous waste is, by definition, potentially dangerous or harmful to human health or the environment. The US Environmental Protection Agency and DEP both have standards for determining when waste materials must be managed as hazardous waste. Hazardous waste can be liquids, solids, or contained gases. It can include manufacturing by-products and discarded used or unused materials such as cleaning fluids (solvents) or pesticides. Hexavalent chromium, a hazardous waste of particular human health concern, is generated as part of the chrome plating process.[20] Used oil and contaminated soil generated from a site clean-up can be hazardous waste. Generators of hazardous waste may treat waste onsite or send it elsewhere for disposal. The potential health effects that come from living near hazardous waste disposal sites have been examined in a number of studies.[21] While there is sometimes limited assessment of exposures that occur in nearby populations, there are studies that have found health effects, including diabetes and cardiovascular disease, associated with living in proximity to hazardous waste sites.[22]

Method: Permitted facility locations were obtained from the MassGIS data layer MassDEP Major Facilities. Facilities were scored on a weighted scale in consideration of the type, permit status, and compliance history for the facility (see Matrix). The weights for all facilities were adjusted based on the distance they fell from populated census blocks. All facilities further than 1,000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for facilities less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given census block group. Facilities outside of a census block group, but less than 1000m from one of that block group's populated blocks were similarly adjusted based on the distance to the nearest block from that block group. Each census block group was scored based on the sum of the adjusted weights for sites it contains or is near.

Data years: 2024

Originating geographic scale of data: point locations

Source: MassDEP Major Facilities from MassGIS at https://www.mass.gov/info-details/massgis-data-massdep-major-facilities.

Weighting Matrix for Permitted Hazardous Waste Facilities and Hazardous Waste Generators

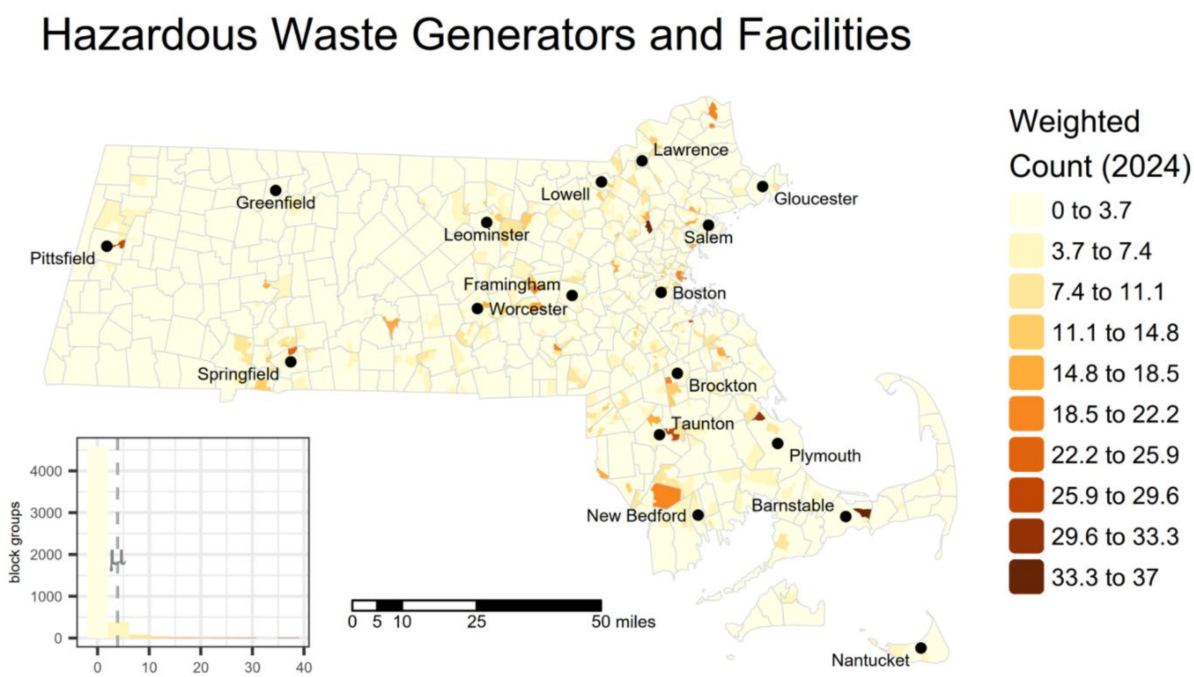
Permitted Hazardous Waste Facilities

Site/Permit Type	Activity or Status	Weight
Facility Activity (base weight)	Hazardous Waste Treatment, Storage and/or Disposal Facility	10
	Hazardous Waste Recycler	7
Permit Type (additional weight)	Large facilities	1
	Air operating permit	1
	RCRA facilities	2

Hazardous Waste Generators

Generator Type	Quantity of Waste	Weight
Large Quantity Hazardous Waste Generators or Large Quantity Toxic User	>1,000 tons/yr	2

Figure 17 - Map of Hazardous Waste Generators and Facilities raw values



Solid Waste Sites and Facilities

Definition: Weighted count of solid waste sites and facilities.

Description: Many newer solid waste landfills are designed to prevent the contamination of air, water, and soil with hazardous materials. However, older sites that are out of compliance with current standards or illegal solid waste sites may degrade environmental conditions in the surrounding area and may expose nearby residents. Other types of facilities, such as composting, treatment and recycling facilities, may raise concerns about odors, vermin, and increased truck traffic. *The presence of a solid waste facility does not by itself mean that there has been any harmful exposure to hazardous pollutants in the surrounding neighborhoods. The potential for exposure, where it may exist, varies widely across facilities.*

Indicator type: *Environmental Effects*

Rationale: Solid waste sites can have multiple impacts on a community. Waste gases like methane and carbon dioxide can be released into the air from disposal sites for decades, even after site closure.[23] Fires, although rare, can pose a health risk from exposure to smoke and ash.[24] Odors and the known presence of solid waste may impair a community's perceived desirability and affect the health and quality of life of nearby residents.[25]

Method: Solid waste site locations were obtained from the MassGIS MassDEP Solid Waste Diversion and Disposal layer. The data includes locations of transfer, incineration and/or land disposal of solid waste in Massachusetts. The weights for all sites, including the large landfill perimeters, were adjusted based on the distance they fell from populated census blocks. Sites further than 1000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for sites less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given census block group. Sites outside of a census block group, but less than 1000m from one of that tract's populated blocks were similarly adjusted based on the distance to the nearest populated block from that block group. Each census block group was scored based on the sum of the adjusted weights for sites it contains or is near. Summed census block group scores were sorted and assigned percentiles based on their position in the distribution.

Data years: 2025

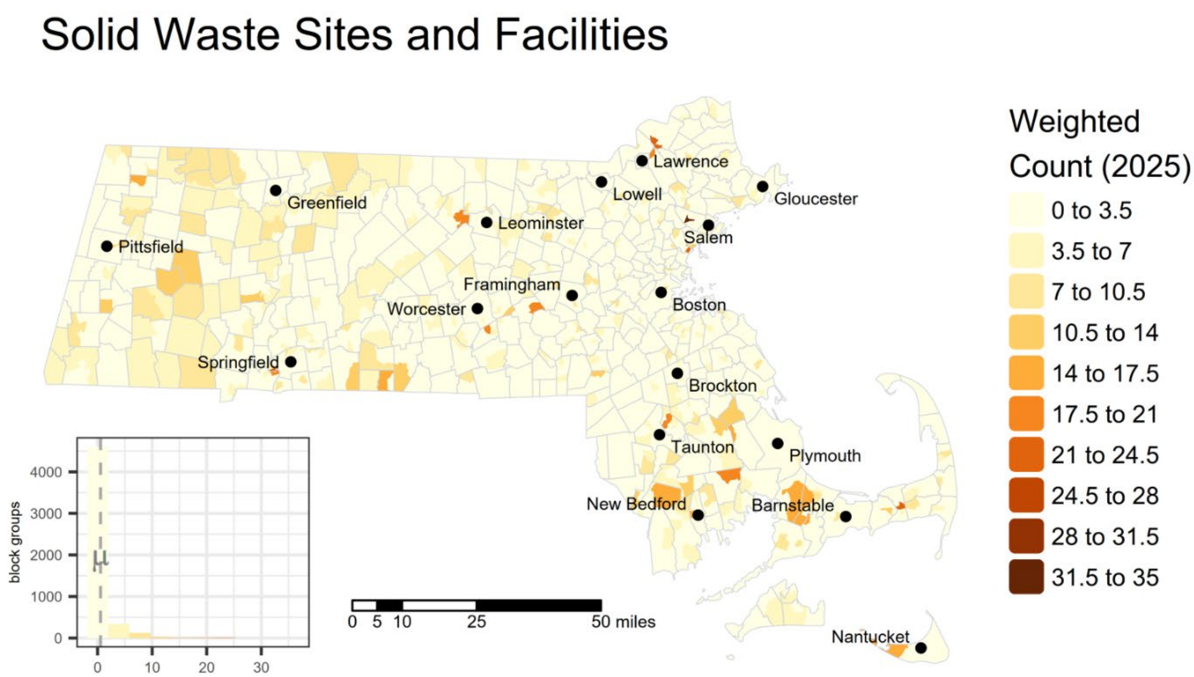
Originating geographic scale of data: point locations

Source: MassDEP Solid Waste Diversion and Disposal layer from MassGIS at https://www.mass.gov/info-details/massgis-data-massdep-solid-waste-diversion-and-disposal.

Weighting Matrix for Solid Waste Sites and Facilities

Category	Site or Facility Type	Weight
Waste combustion facility (active)		10
Solid Waste Landfill or Construction, Demolition and Inert (CDI) Debris Waste Disposal (active)		8
Unpermitted Dumping Grounds (inactive)		6
Transfer, processing, handling facility	Permitted large volume, C&D	5
	Permitted small volume, organic or wood waste	2
Waste Tire, recycling, wood waste		4
Composting	Site assigned composting facility	4
	Site assignment exempt composting facility	2
Aerobic/Anaerobic Digesting		3
Solid Waste Disposal Site, historic incinerator or combustion facility (closed, inactive)		1

Figure 18 – Map of Solid Waste Sites and Facilities raw values



Impaired Water Bodies

Definition: Count of pollutants across all water bodies designated as impaired within the area.

Description: The Massachusetts Surface Water Quality Standards define the water quality goals for surface waters by designating the most sensitive uses; prescribe minimum water quality criteria (both numeric and narrative) required to sustain the designated uses; and include provisions to maintain and protect existing uses and high-quality waters. Contamination of streams, rivers, lakes, and coastal waters by pollutants can compromise the use of the water body for drinking, swimming, fishing, aquatic life protection, and other designated uses. When this occurs, such water bodies are considered “impaired.” Information on impairments to these water bodies can help determine the extent of environmental degradation within an area.

Indicator type: *Environmental Effects*

Rationale: Rivers, lakes, ponds, estuaries and marine waters in Massachusetts are important for many different uses. Swimming and boating uses can be impacted by high levels of water-borne pathogens and other contaminants. Waterbodies used for recreation may also be important to the quality of life of nearby residents if subsistence fishing is critical to their livelihood. Water bodies also support abundant flora and fauna. Alterations in natural conditions in aquatic environments (e.g., excess nutrients like phosphorous and/or nitrogen, toxics, etc.) can affect biological diversity and overall health of ecosystems. Aquatic species important to local economies may be impacted if the habitats where they seek food and reproduce are changed. Marine wildlife like fish and shellfish that are exposed to toxic substances may potentially expose local consumers to toxic substances as well. Excessive hardness, unpleasant odor or taste, turbidity, color, weeds, and trash in the waters are types of pollutants affecting water aesthetics, which in turn can affect nearby communities.

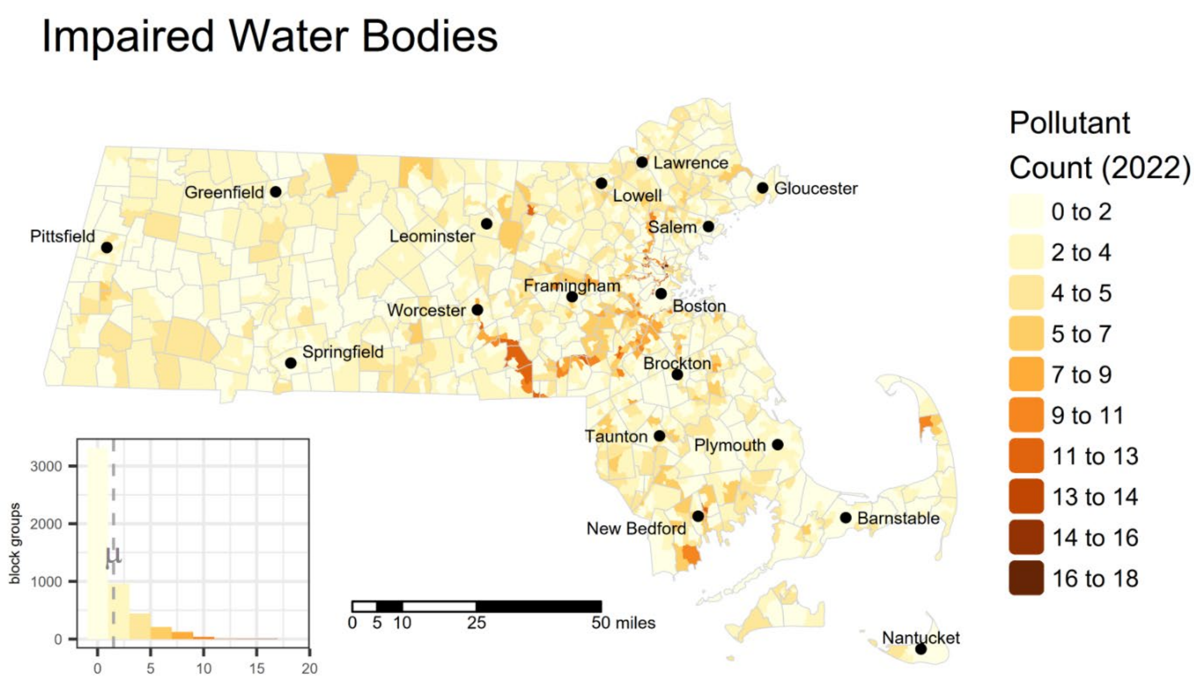
Method: The number of pollutants listed in streams or rivers that fell within 1 kilometer (km) or 2 km respectively of a census block group’s populated blocks were counted. The 2 km buffer distance was applied to major rivers (>100 km in length). The 1 km buffer distance was applied to all smaller streams/rivers. The number of pollutants listed in lakes, bays, estuaries or shoreline that fell within 1 km or 2 km of a census block group’s populated blocks were counted. The 2 km buffer distance was applied to major lakes or bays greater than 25 square kilometers in size. The 1 km buffer distance was applied for all other lakes/bays. The two pollutant counts were summed for every census block group. Each census block group was scored based on the sum of the number of individual pollutants found within and/or bordering it. For example, if two stream sections within a census block group were both listed for the same pollutant, the pollutant was only counted once. Summed census block group scores were sorted and assigned percentiles based on their position in the distribution.

Data years: 2022

Originating geographic scale of data: water body linear and polygon features

Source: MassDEP 2022 Integrated List of Waters (305(b)/303(d)) from MassGIS at https://www.mass.gov/info-details/massgis-data-massdep-2022-integrated-list-of-waters-305b303d.

Figure 19 - Map of Impaired Water Bodies raw values



Pollution and Climate Burden - Climate Risks

Drought

Definition: Sum of weekly total percent of an area experiencing a severe, extreme, or exceptional drought (categories D2, D3, or D4), adapted from Colorado EnviroScreen.
Description: A drought is a period of time when an area experiences less rainfall or snowfall than normal. Climate change increases the frequency and severity of drought. This indicator represents how much of a county was affected by drought over time. These levels of drought imply some level of voluntary or mandated water use restrictions and observable damage or loss of pasture and crops. The U.S. Drought Monitor reports the percentage of each county experiencing each of the six drought levels (None, D0, D1, D2, D3, and D4) each week. The impact of drought varies significantly from community to community based on land use patterns and water supply storage and availability. Efficiency requirements are applied where possible to promote the wise use of our water resources.

Indicator type: *Climate Risks*

Rationale: Drought is an important climate health indicator because it reflects areas with an increased risk of wildfires, dust storms, shortage of local water and food sources, cyanobacterial blooms, poor quality drinking water, and sanitation and hygiene restrictions. Drought can lead to dehydration, stress, anxiety, depression, lung and heart diseases, stroke, and infectious diseases, among other impacts.[26] Severe, extreme, and exceptional drought categories are more likely to impact populations through water shortages, water restrictions, or crop or pasture losses.

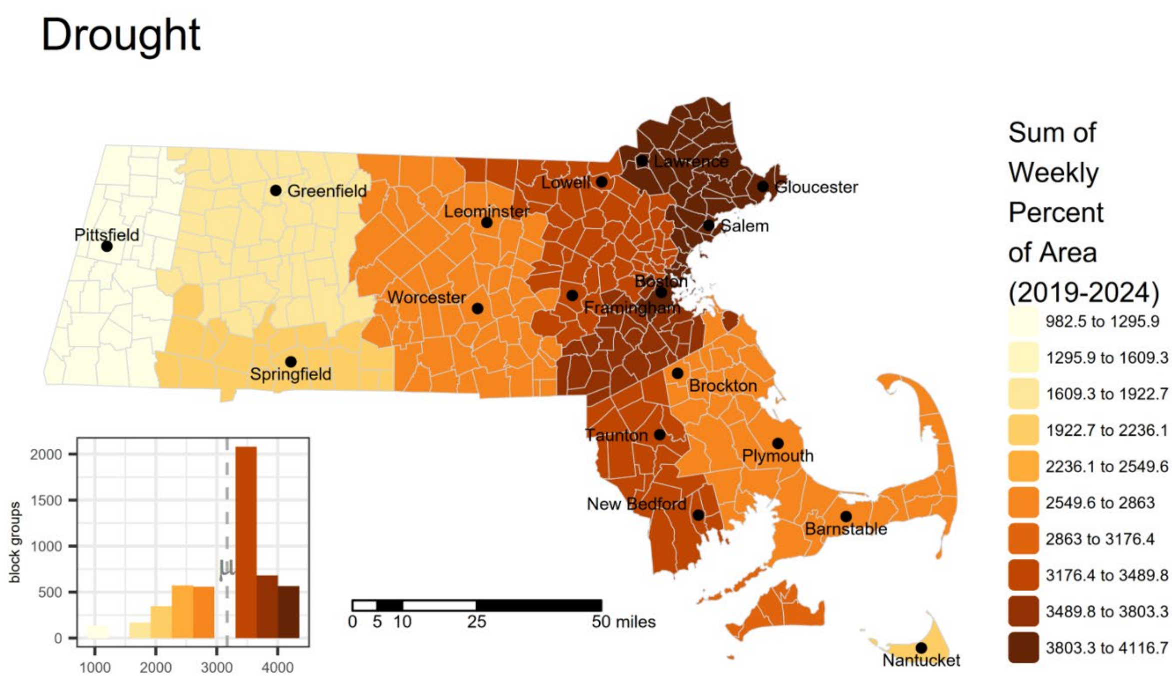
Method: The U.S. Drought Monitor reports the percentage of each county experiencing each of the six drought levels (None, D0, D1, D2, D3, and D4) each week. The sum of areas experiencing D2, D3, D4 level droughts was calculated weekly across all weeks from January 2019 to December 2024. The sum of the weekly drought values across that time period was used to define the Drought measure. All census block groups received the Drought value for the county in which they are located. The Massachusetts Wetlands Protection regulations for the Riverfront Area at 310 CMR 10.58(2)(a)1.f., and the definition of a Pond at 310 CMR 10.04, don't allow observations of River flow and Pond/Lake area during the specific periods of time when the Massachusetts Secretary of Energy and Environmental Affairs declares there is an Advisory or worse Drought (Level 1 to Level 4 Drought).

Data years: 2019 - 2024

Originating geographic scale of data: county

Source: U.S. Drought Monitor 2019-2024 https://droughtmonitor.unl.edu/Data.aspx

Figure 20 - Map of Drought index raw values



Wildfire Risk

Definition: Mean wildfire hazard potential.

Description: A wildfire is an uncontrolled fire in nature, such as in forests or grasslands. Wildfires affect people's physical, mental, and financial health. Wildfires threaten ecosystems and water supplies and produce harmful smoke, which can trigger health problems like asthma or stroke. Climate change increases the risk and severity of wildfires.

Indicator type: *Climate Risks*

Rationale: Wildfires can lead to direct injuries and death among those living in affected areas.[27] Other health risks can be associated with the evacuation of patients, reduced access to health services, and the impact on the quality and availability of local water resources. Wildfires also produce smoke that impacts communities in the affected areas. The smoke can travel far distances, even reaching communities in other states. Wildfire smoke has been associated with multiple health outcomes, such as adverse pregnancy outcomes, impacts to children's health (e.g., asthma), and impacts to adults (e.g., cardiovascular and respiratory diseases). Exposure to wildfire smoke can ultimately increase the risk of premature mortality.[28]

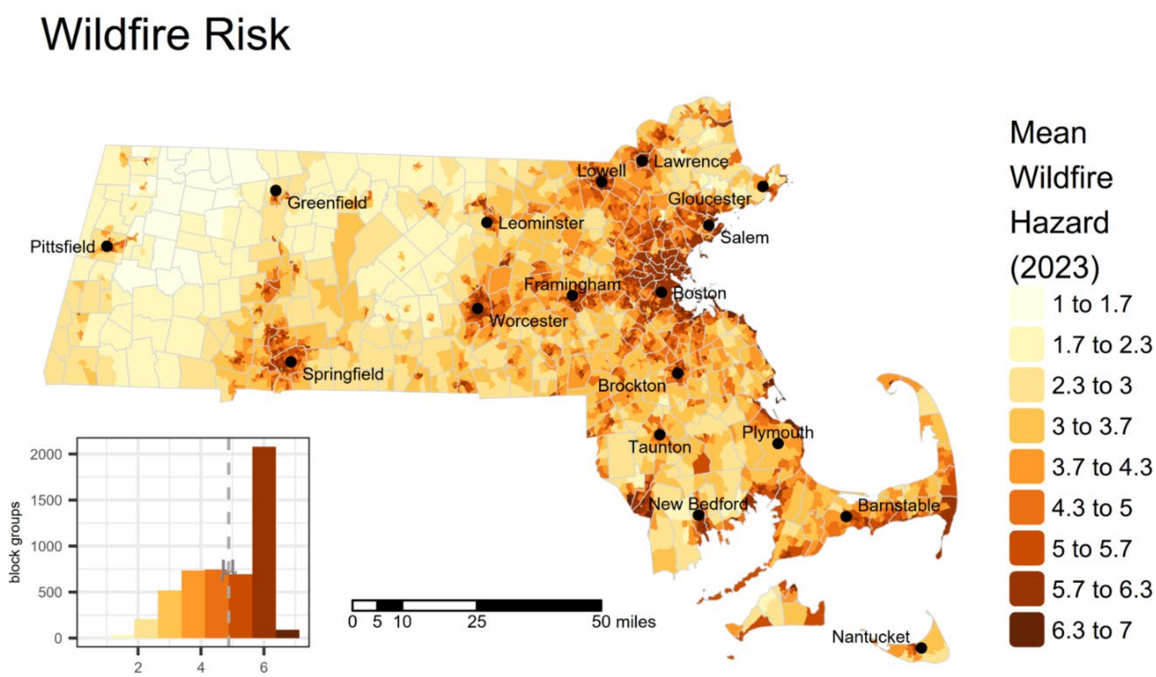
Method: The mean wildfire hazard potential within each geographic area is used as the Wildfire risk score. Wildfire Hazard Potential (WHP) is provided as a 270-meter resolution raster. The mean WHP of all pixels or raster grid cells falling within each census block group were calculated.

Data years: 2023

Originating geographic scale of data: 270-meter pixels

Source: U.S. Department of Agriculture (USDA), U.S. Forest Service (USFS)
https://www.fs.usda.gov/rds/archive/catalog/RDS-2015-0047-4

Figure 21 - Map of Wildfire Risk index raw values



Flood Risk

Definition: Percentage of each geographic area where there is at least a one-percent chance of flooding annually.

Description: Special Flood Hazard Areas (SFHAs) are defined by the Federal Emergency Management Agency as the area that will be inundated by the flood event having a 1-percent chance of being equaled or exceeded in any given year.[29] The 1-percent annual chance flood is also referred to as the base flood or 100-year flood and is considered to be an area with a high risk of flooding.[30] The 100-year flood is significant in the Massachusetts Wetlands Protection Act (WPA) because it is used as a regulatory threshold to identify and protect floodplain areas that are critical to public safety, property protection, and ecosystem health. Under the WPA, areas that are subject to inundation by the 100-year flood (as defined by FEMA or calculated hydrologically) are classified as Bordering Land Subject to Flooding, which is a protected wetland resource area. Activities proposed in these areas are subject to wetlands permitting and review to ensure they do not cause adverse impacts like increased flooding, displacement of floodwaters, or loss of flood storage.

Indicator type: *Climate Risks*

Rationale: Floods are the most common form of weather-related disaster in the US, and they are the second-leading cause of weather-related fatalities.[31] Floods can immediately impact health due to drowning, injuries, and hypothermia. Other health risks can be associated with the evacuation of patients, reduced access to health services, or the impact on basic services, including drinking water supply and quality. In the medium and long term, floods have been associated with infectious diseases (due to poor quality drinking water, lack of hygiene and sanitation, infected wounds, or insect bites), poisoning (due to chemicals in floodwater), and mental health issues (e.g., stress, anxiety, depression). Flood exposure can ultimately increase the risk of premature mortality.[32] Climate change increases the risk and severity of floods.

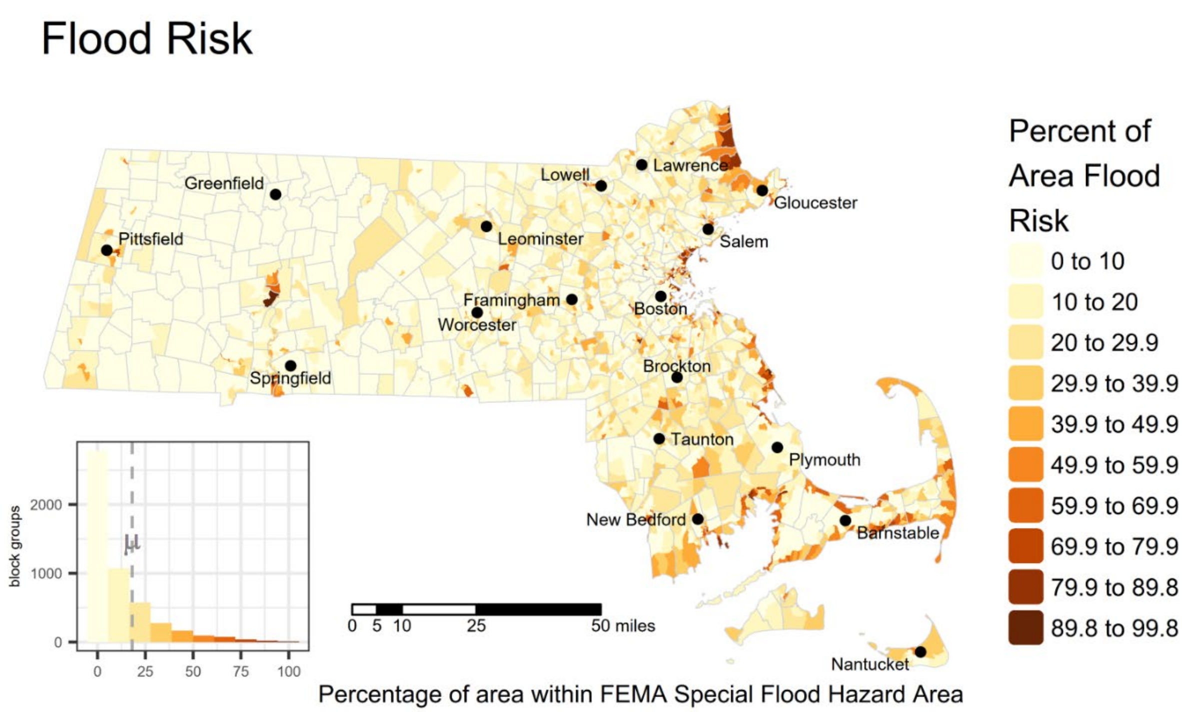
Method: FEMA National Flood Hazard Layer (NFHL) digital Flood Insurance Rate Maps (DFIRMs) were downloaded from MassGIS. However, as of July 2025, NFHL DFIRMs are not available for the northwest quadrant of Massachusetts – Berkshire County, Hampshire County, Franklin County, and portions of northern Worcester County. These areas were supplemented with data from the FEMA Q3 Flood Zones from Paper FIRMs layer from MassGIS, which cover most of the northwest except for Franklin County. Flood risk data for Franklin County was acquired from FEMA's National Risk Index – "riverine flooding - exposure - impacted area" layer, which is the percentage of area exposed to flood risk, and is summarized by census tract. For NFHL DFIRMs and FEMA Q3 layers, the area of all features within the 1% Annual Chance Flood Hazard within each census block group was divided by the total area of the census block group. For NFHL DFIRMs, these included areas within zones "A", "AE", "AH", "AO", and "VE". For FEMA Q3 data, these included areas within zones "AE", "A", "D", and "AO". If no flood areas were found within the census block group, a value of zero was used. For FEMA NRI data in Franklin County, all census block groups were assigned the value of the census tract in which they are contained.

Data years: FEMA NFHL data 2023; FEMA Q3 Flood Zones from Paper FIRMs data1997 - 2013; FEMA National Risk Index data for Franklin County 2025

Originating geographic scale of data: delineated flood polygons for NHFL and Q3 data; census tracts for Franklin County

Source: FEMA National Flood Hazard Layer (NFHL) from MassGIS at https://www.mass.gov/info-details/massgis-data-fema-national-flood-hazard-layer; FEMA Q3 Flood Zones from Paper FIRMs from MassGIS at https://www.mass.gov/info-details/massgis-data-fema-q3-flood-zones-from-paper-firms; FEMA's National Risk Index – "riverine flooding - exposure - impacted area" layer from https://hazards.fema.gov/nri/data-resources

Figure 22 - Map of Flood Risk raw values



Extreme Heat Days

Definition: Number of days between May and September from 2015 through 2024 in which daily high temperature was 85 degrees Fahrenheit or higher.

Description: Extremely hot days can cause health risks due to heat exhaustion and dehydration. Extremely hot days can also cause electrical outages, which can hurt the health of people who use machines for healthcare purposes. Climate change increases the number and severity of extreme heat days. This indicator describes the estimated total number of days from May to September for the period 2015 – 2024 that were 85 degrees Fahrenheit or higher, a temperature threshold associated with a significant increase in the rate of heat-related emergency department visits according to the Massachusetts Department of Public Health.

Indicator type: *Climate Risks*

Rationale: Heat-related illnesses occur when people are exposed to an abnormal or prolonged amount of heat without relief or adequate fluid intake.[33] Heat-related illnesses include heat stroke, exhaustion, cramps, rash, and sunburn. Heat-related illnesses can also increase the risk of premature mortality. Infants, children, older adults, and those living with medical conditions are more susceptible to developing health-related issues. Moreover, outdoor workers, athletes, and those without access to shelter, air conditioning, and drinking water are also more vulnerable to extreme heat days.[34]

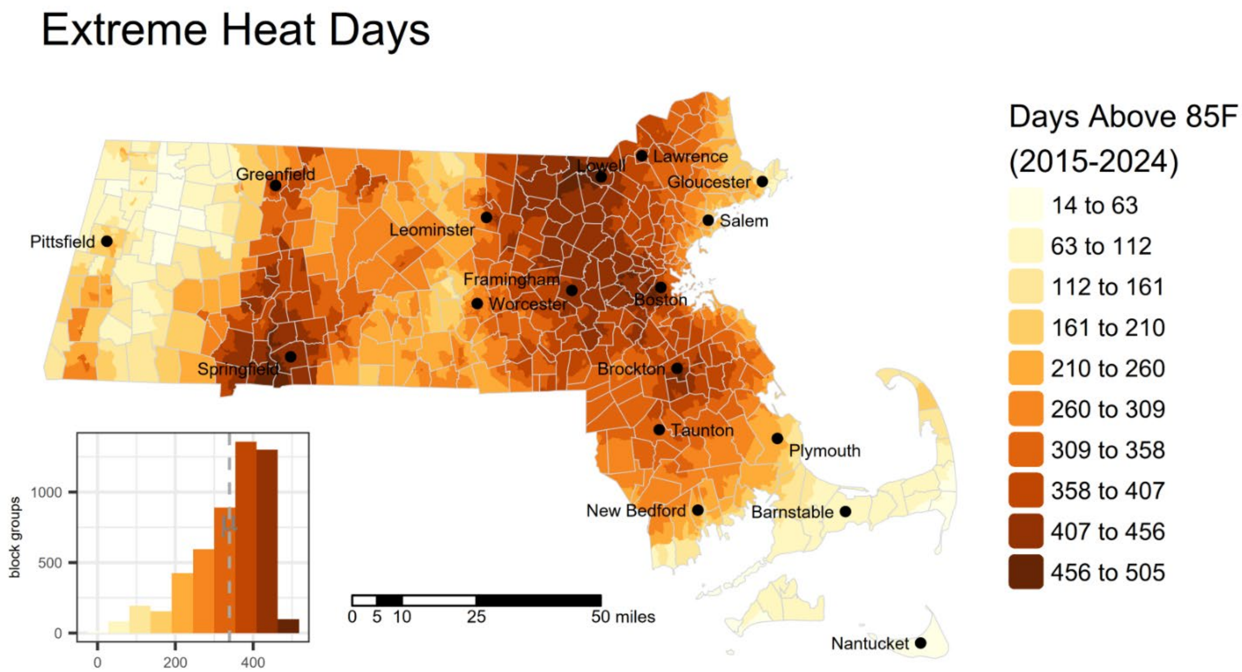
Method: Data on daily maximum temperatures was downloaded from the Oregon State University PRISM Group via the "prism" package in R. Datasets are provided as single-layer grids covering the coterminous US for each day at 800-meter resolution. These grids were stacked and summed, providing a total count of the number of days at or above 85F per pixel or grid cell. Each census block group in Massachusetts was assigned the average count of overlapping grid cells.

Data years: 2015 – 2024

Originating geographic scale of data: 800m pixels or grid cells

Source: PRISM Group, Oregon State University, https://prism.oregonstate.edu, data created November 2015 through April 2025, accessed 9 Aug 2025.

Figure 23 - Map of Extreme Heat Days raw values



Population Characteristics - Sensitive populations

Pediatric Asthma

Definition: Population-weighted average asthma prevalence (percentage of K-8 enrollment). The rate in Massachusetts is 10.5%.

Description: Asthma prevalence in schools is defined as the percentage of enrolled students reported by school nurses to have asthma during a school year. Schools are public, charter, and private. School-based asthma data are reported annually by school nurses through the Massachusetts Department of Public Health Pediatric Asthma and Diabetes Survey for students in grades K-8 who have ever been diagnosed with asthma. Average prevalence for each census block group was calculated based on the average prevalence of all schools associated with that block group, weighted by school enrollment.

Indicator type: *Sensitive Populations*

Rationale: Asthma is an illness that affects the respiratory tract and airways that carry oxygen into and out of the lungs. During an asthma attack, these airways constrict, resulting in wheezing and difficulty breathing. Asthma can affect people of all ages. However, it often starts in childhood and is more common in children than adults. Asthma is a common chronic disease that continues to increase in prevalence. It is the most common chronic disease in children. The state of Massachusetts has an elevated rate of asthma compared to the national prevalence rate.[35] Causes of asthma are unknown. However, episodes of asthma (asthma attacks) can be triggered by certain environmental pollutants such as air pollution, mold, pets/pet dander, and dust mites. A number of studies have reported links between exposure to air pollution and asthma. Several criteria air pollutants are known to exacerbate asthma symptoms and contribute to asthma development. In addition, climate change may affect air quality and worsen asthma in several ways. Climate-driven changes in weather can increase certain criteria air pollutants like ozone and particulate matter, as well as other triggers such as pollen, mold, dust mites, and bacteria, all of which may exacerbate asthma. Climate change also leads to more frequent wildfires, which worsen respiratory illnesses including asthma. Reducing exposure to these pollutants can help prevent symptoms. Asthma impacts certain groups more than others, including children, older adults, people of color, and those of lower socioeconomic status. These disparate impacts may be due to environmental factors related to socioeconomic status, like housing conditions or proximity to sources of pollution. Children of color have a higher prevalence of asthma and asthma-related emergencies. Children with asthma seem to be impacted by particle pollution more so than adults with asthma.[36]

Method: MDPH calculated the school rates for asthma prevalence and provided these to MassDEP with values per school. Each school was associated with census block groups using the following steps:

1. Data Source: The public schools dataset is from [MassGIS - Massachusetts Schools (Pre-K through High School)], dated December 2024.
2. Initial Coverage: School districts, towns, and ZIP codes were assigned to block groups based on centroid location to ensure full statewide coverage.
3. District-Level Assignment: If a school is the only one of its type in a district, all block groups in that district are assigned to it.
4. Town-Level Assignment: For unassigned cases, if only one school of that type exists in a town, its block groups are assigned to that school.
5. ZIP Code-Level Assignment: If still unassigned, block groups are matched to a school by ZIP code if only one school of that type exists in the ZIP.
6. 0.5-Mile Buffer: Remaining unassigned block groups are matched to schools within a 0.5-mile buffer, limited to schools in the same district.
7. Fallback Matching: For all types except Other/Unknown, any remaining block groups are assigned to the *nearest* school of the same type within the same district.

The pediatric asthma data is based on the original CIA dataset from MDPH (2017–2024). Please note that 15 block groups remain unmatched due to suppressed or missing data (e.g., missing enrollment counts or closed schools with outdated codes).

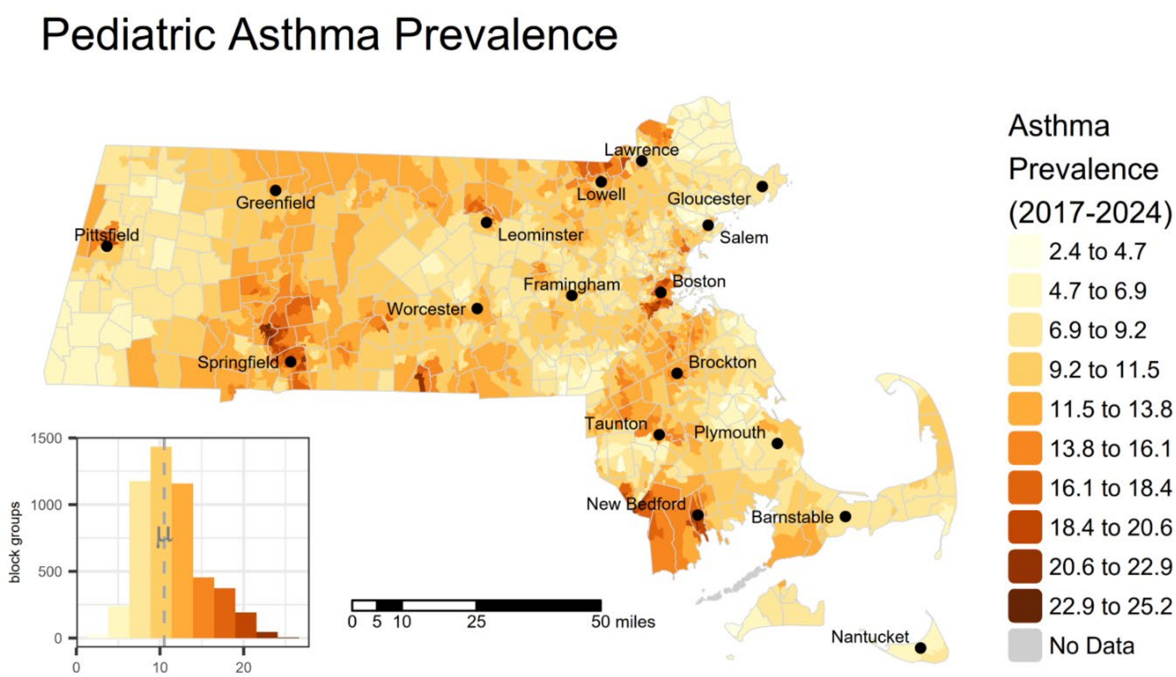
Average prevalence for each census block group was calculated based on the average prevalence of all schools associated with each block group, weighted by school enrollment. Schools missing asthma or enrollment data were excluded.

Data years: 2017 – 2024

Originating geographic scale of data: school point locations

Source: MassDEP Cumulative Impact Analysis in Air Quality Permitting at https://www.mass.gov/info-details/cumulative-impact-analysis-in-air-quality-permitting#cia-guidance-and-tools-

Figure 24 - Map of Pediatric Asthma Prevalence raw values



Elevated Blood Lead Levels in Children

Definition: 5-year average prevalence of elevated (≥5 µg/dL estimated confirmed) childhood blood lead levels in children (ages 9-47 months). The prevalence in Massachusetts is 18.4 per 1,000 children screened.

Description: Exposure to lead through paint is the most significant source of lead exposure for children. Lead is a toxic heavy metal and occurs naturally in the environment. However, most of the high levels of lead found in our environment result from human activities. Historically, lead was used as an additive in gasoline and as a primary ingredient in house paint. Lead levels in the United States have declined over the past five decades due to various regulations. However, lead still persists in older buildings containing lead paint, as well as old plumbing and contaminated soil. Young children absorb lead more easily than adults. The Massachusetts Lead Poisoning Prevention and Control Act is a state law that requires all children to be screened each year for lead poisoning through age three. Children living in high-risk communities must be screened each year through age four. All children must show proof of screening at least once to enter daycare, pre-kindergarten programs, and kindergarten.

Indicator type: *Sensitive Populations*

Rationale: Young children are especially susceptible to the effects of lead exposure and can suffer profound and permanent adverse health effects, particularly in the brain and nervous system.[37] This increased susceptibility is due to their unique exposure pathways (e.g., dust-to-hand-to-mouth), developing brains, and differences in the absorption of ingested lead.[38] Lead in the body can: hurt the brain, kidneys, and nervous system, slow down growth and development, make it hard to learn, damage hearing and speech, and cause behavior problems. There are no known safe levels of lead exposure, and levels that were previously considered safe are now known to cause subtle, chronic health effects. In October 2021, the Centers for Disease Control and Prevention (CDC) lowered the blood lead reference value (BLRV) from 5 micrograms per deciliter (µg/dL) to 3.5 µg/dL.

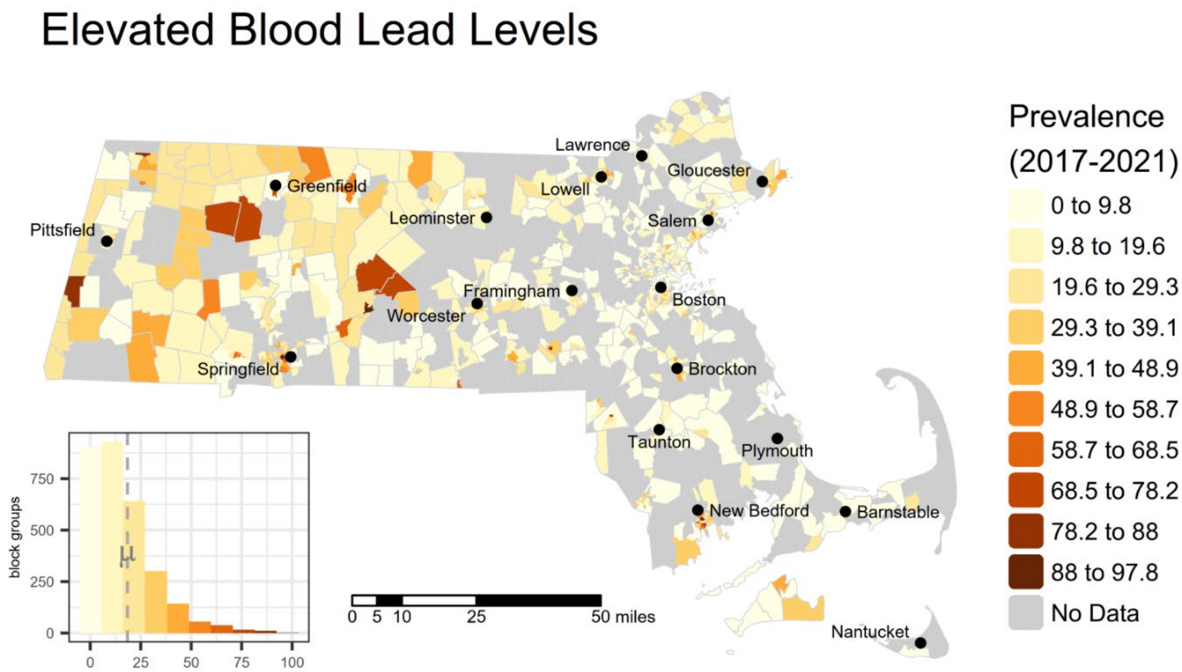
Method: All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2017 – 2021

Originating geographic scale of data: census tracts

Source: Massachusetts Environmental Public Health Tracking via MassDEP Cumulative Impact Analysis in Air Quality Permitting at https://www.mass.gov/info-details/cumulative-impact-analysis-in-air-quality-permitting#cia-guidance-and-tools-

Figure 25 - Map of Elevated Blood Lead Levels raw values



Low Birth Weight Infants

Definition: 5-Year annual average low birth weight (less than 2,500 grams) per 100 full-term births. The state rate is 2.2 per 100 full-term births.

Description: Birth weight, which is impacted by the length of gestation and fetal growth, is considered low when infants weigh less than 2,500 grams (5lbs and 8 ounces). Preterm birth and low birth weight are associated with a range of health issues for children, including breathing problems, feeding difficulties, developmental delay, hearing and vision problems, and more.

Indicator type: *Sensitive Populations*

Rationale: Compared to normal birth weight infants, low birth weight infants may be more at risk for illness through the first six days of life, infections, and long-term impairment, such as delayed motor and social development or learning disabilities. Preterm birth and low birth weight are associated with a range of health issues for children, including breathing problems, feeding difficulties, developmental delay, hearing and vision problems, and more.[39] There are many potential causes for adverse birth outcomes, including maternal health conditions and lifestyle factors.[40] The role that exposure to environmental contaminants plays in adverse birth outcomes is not yet fully understood, particularly in the potential causation of preterm birth and low birth weight.[41] Various exposures have been implicated as risk factors for full-term low birth weight, including maternal exposure to lead and exposure to toxic substances in air, water or food.[42] Research suggests that exposure to PM2.5, carbon monoxide, and nitrogen oxides might contribute to preterm birth and low birth weight.[43] Exposure to lead is also known to cause reduced fetal growth and may cause preterm birth.[44] Preterm and low birth weight infants have a higher risk of health issues and mortality from childhood to adulthood.[45]

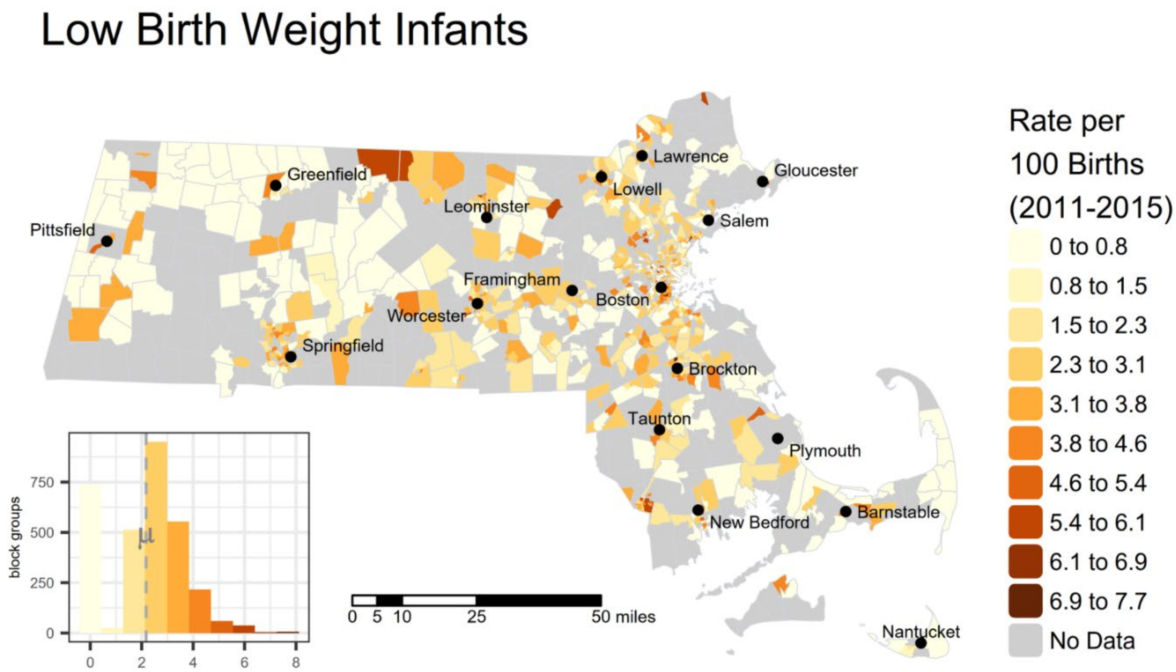
Method: All census block groups received the prevalence value for the census tract in which they are located. Some block groups have no data due to very small numbers that are suppressed by MDPH to protect privacy.

Data years: 2011-2015

Originating geographic scale of data: census tract

Source: Massachusetts Department of Public Health Birth Outcomes Data of Massachusetts Residents at https://www.mass.gov/info-details/birth-outcomes-data-of-massachusetts-residents via MassDEP Cumulative Impact Analysis in Air Quality Permitting at https://www.mass.gov/info-details/cumulative-impact-analysis-in-air-quality-permitting#cia-guidance-and-tools-

Figure 26 - Map of Low Birth Weight Infants raw values



Premature Mortality (PM)

Definition: Age-adjusted premature mortality rate (per 100,000). The rate in Massachusetts is 292.8 per 100,000 residents.

Description: Premature mortality tracks unfulfilled life expectancy and is based on deaths prior to age 75. Premature mortality includes unintentional injuries such as motor vehicle-related deaths, poisonings, falls, fires, and drownings that were not intended to occur. Deaths are for all age groups.

Indicator type: *Sensitive Populations*

Rationale: Premature mortality is a powerful marker of multiple diseases and the health status in a community. Life expectancy can be improved in communities through public health and clinical care practices. Reduced life expectancy has been associated with environmental risk factors, such as air, water, and soil pollution, noise, proximity to traffic and contaminated sites, and climate risks (droughts, floods, wildfires, and extreme heat days), among others.

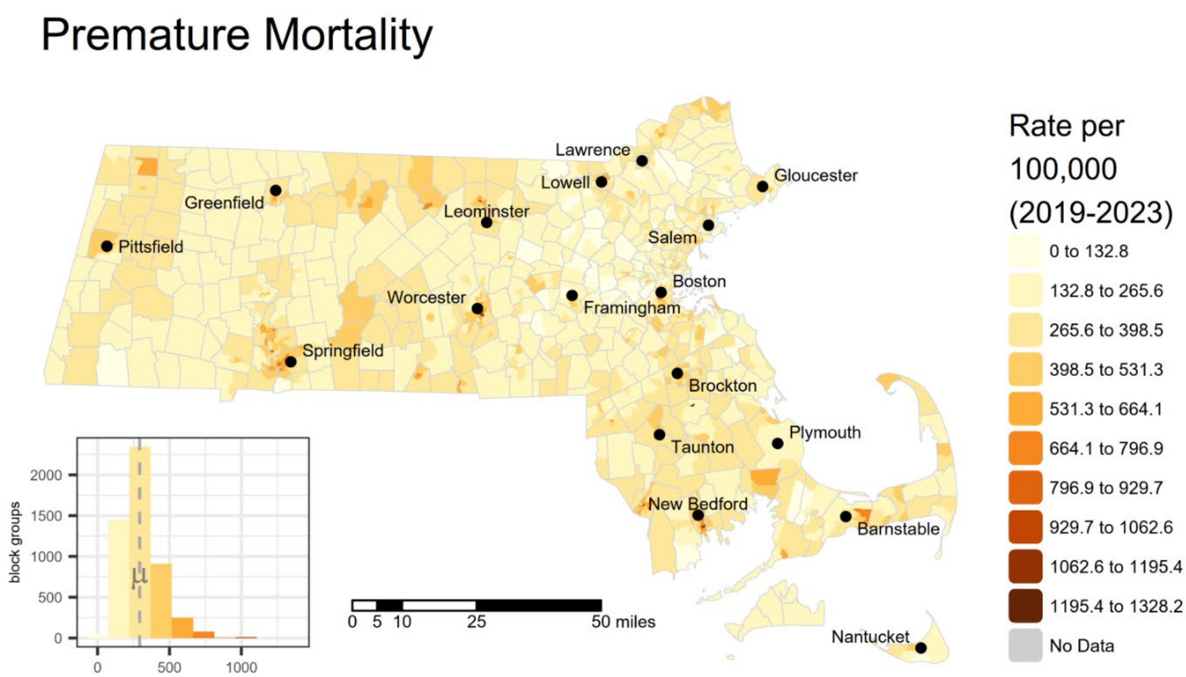
Method: Massachusetts Department of Public Health (MDPH) provided an age-adjusted premature mortality rate (PMR per 100,000) by census tract.[46] All census block groups received the mortality rate for the census tract in which they are located.

Data years: 2019 – 2023

Originating geographic scale of data: census tracts

Source: Massachusetts Department of Public Health Death Data Registry of Vital Records and Statistics acquired via MassDEP Cumulative Impact Analysis in Air Quality Permitting at https://www.mass.gov/info-details/cumulative-impact-analysis-in-air-quality-permitting#cia-guidance-and-tools-

Figure 27 - Map of Premature Mortality raw values



Adult High Blood Pressure

Definition: Prevalence of high blood pressure among adults. The prevalence in Massachusetts is 29.9%.

Description: High blood pressure, also called hypertension, is blood pressure that is higher than normal. Higher blood pressure levels put individuals at greater risk for other health problems, such as heart disease, heart attack, and stroke.

Indicator type: *Sensitive Populations*

Rationale: Hypertension, a type of cardiovascular disease (CVD) that leads to a prolonged increase in blood pressure, is a major risk factor for other CVD conditions like coronary heart disease and stroke. Risk factors for hypertension include obesity, physical inactivity, and high sodium consumption.[47] Traditional risk factors for CVD, like male sex, older age, increased blood pressure, high cholesterol, and smoking, account for about 50 percent of cardiac events. There are also known environmental exposures that act independently or in conjunction with established risk factors to impact CVD and hypertension.[48] Lead exposure can lead to both cardiovascular effects and cardiovascular-related mortality. There is consistent evidence to show that lead exposure increases blood pressure in adults, hypertension, and cardiovascular mortality, with more limited information on lead relating to changes in heart rate variability and the development of atherosclerosis.[49] Air pollution exposure has also been found to contribute to the development of CVD and hypertension and exacerbate existing CVD conditions. Evidence is particularly strong for short-term and long-term exposures to fine particulate matter, or particulate matter with a diameter of less than 2.5 μm.[50] In addition, the effects of climate change may cause or worsen certain illnesses and health conditions. Climate-related changes to weather and heat can worsen air quality by impacting ozone and particulate matter concentrations, wildfires, and allergens. The effects of climate change on air quality pose human health risks, including heart disease and stroke. People with conditions such as hypertension are particularly vulnerable to the adverse effects of climate change on outdoor air quality. Certain populations remain at a disproportionate risk for exposure and negative health outcomes. Those at increased risk of exposure to particle pollution include non-white and low-income populations, as well as those who live or work in urban and industrial areas. Elderly and other high-risk populations are also heavily impacted. Studies have shown exposure to ambient airborne particulate matter could be associated with increased hospitalizations and mortality among older individuals, largely due to cardiopulmonary and cardiovascular disease.[51]

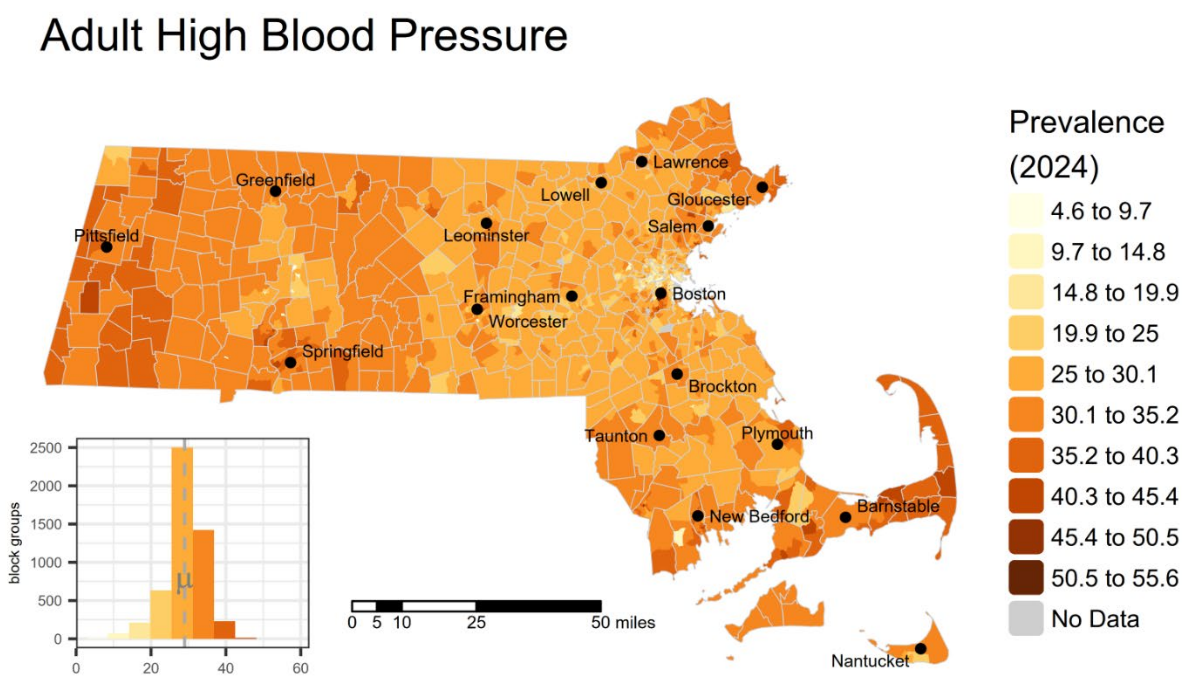
Method: Data provided a census tract level. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2024

Originating geographic scale of data: census tracts

Source: CDC PLACES Health Outcomes at https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data

Figure 28 - Map of Adult High Blood Pressure raw values



Coronary Heart Disease

Definition: Prevalence of coronary heart disease among adults. The prevalence in Massachusetts is 4.6%.

Description: Coronary heart disease (also called ischemic heart disease) is caused by plaque buildup on the arteries. These blockages can limit the amount blood and oxygen the heart receives causing chest pain or angina. Coronary heart disease may present as an acute myocardial infarction which happens when the blood flow to a part of the heart is blocked by plaque.

Indicator type: *Sensitive Populations*

Rationale: Coronary artery disease (CAD) is the most common type of heart disease in the United States. Overweight, physical inactivity, unhealthy eating, and smoking tobacco are behavioral risk factors for CAD. There is a growing body of evidence showing that physicochemical factors in the environment contribute significantly to the cardiovascular diseases such as coronary heart disease. Urbanization is often associated with accumulation and intensification of these stressors.[52] Globally, some research concludes that environmental causes may be a larger cardiovascular mortality risk factor than those attributable to metabolic, tobacco use, and behavioral risk factors.[53]

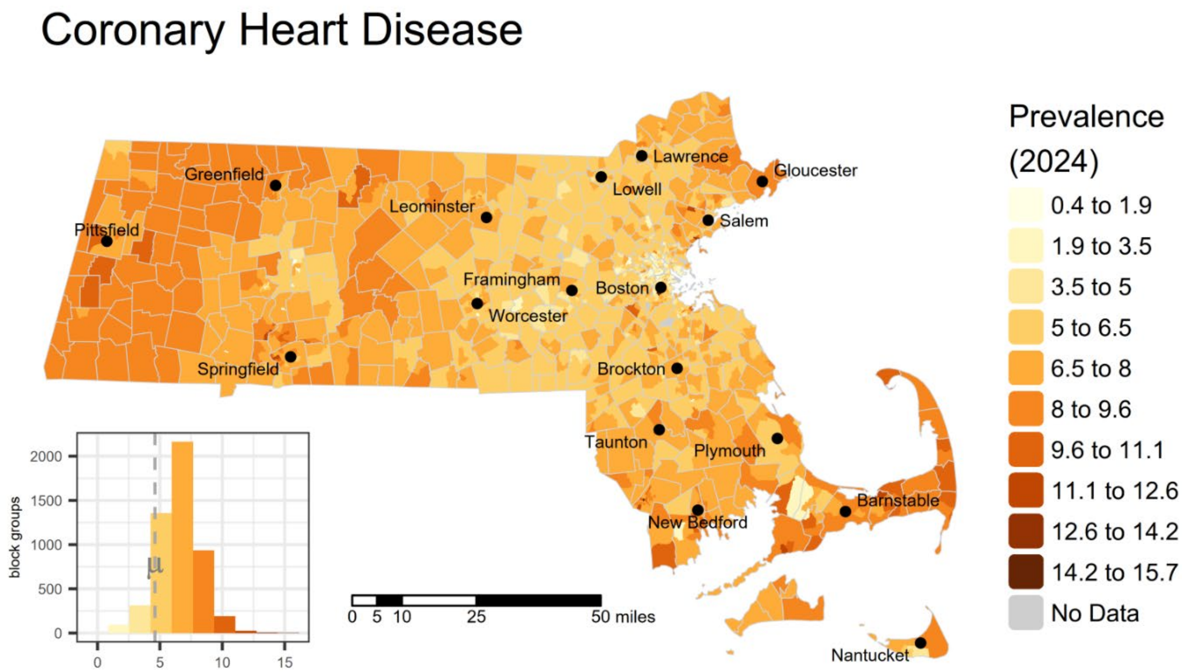
Method: Data provided a census tract level. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2024

Originating geographic scale of data: census tracts

Source: CDC PLACES Health Outcomes at https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data

Figure 29 - Map of Coronary Heart Disease raw values



Chronic Obstructive Pulmonary Disease (COPD)

Definition: Prevalence of chronic obstructive pulmonary disease among adults. The prevalence in Massachusetts is 5.8%.

Description: Chronic Obstructive Pulmonary Disease, or COPD, refers to a group of diseases including emphysema and chronic bronchitis, that block airflow and cause breathing-related problems.

Indicator type: *Sensitive Populations*

Rationale: COPD is a leading cause of death in the US and in Massachusetts. The main cause of COPD is smoking, but nonsmokers can also get COPD. Approximately 85-90% of COPD diagnoses are attributed to cigarette smoking. Occupation exposures may account for another 15% of COPD diagnoses. A history of asthma may increase the risk of developing COPD. Long-term exposure to air pollution, secondhand smoke, dust, fumes and chemicals (which are often work-related) can cause COPD.[54] Although the primary cause of COPD is smoking, an increasing number of studies have reported associations between indoor and outdoor air pollution exposures and COPD, suggesting that environmental exposure could be driving a large percentage of COPD cases.

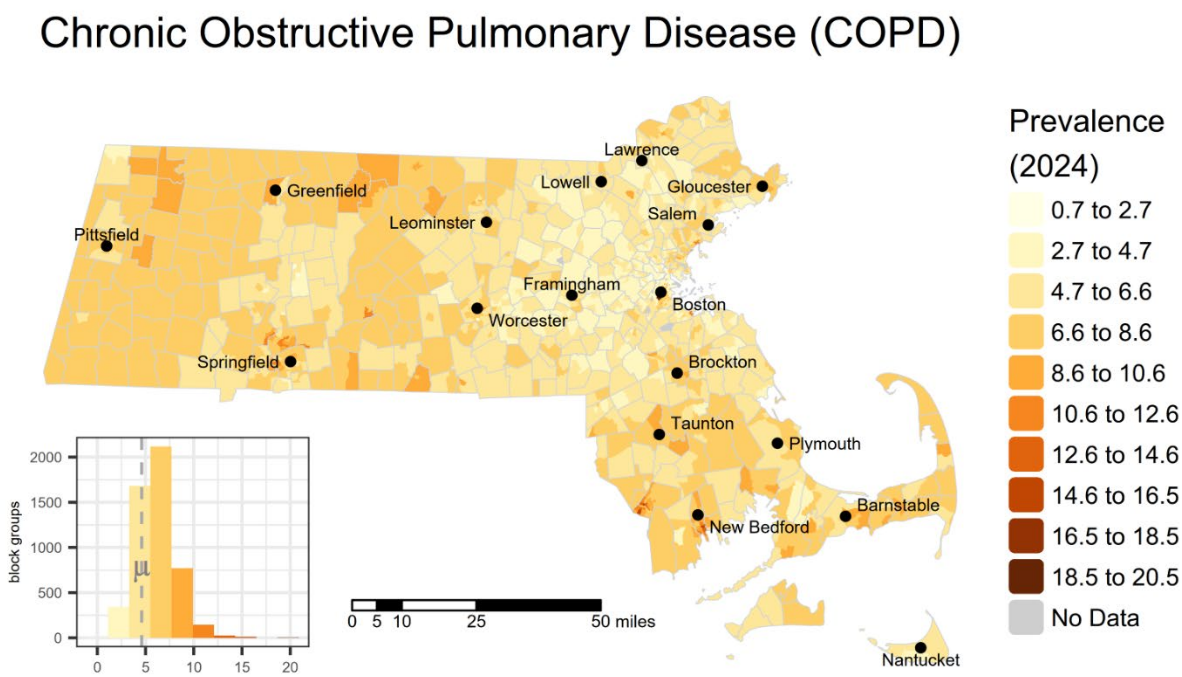
Method: Data provided a census tract level. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2024

Originating geographic scale of data: census tracts

Source: CDC PLACES Health Outcomes at https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data

Figure 30 - Map of Chronic Obstructive Pulmonary Disease (COPD) raw values



Adult Cancer

Definition: Prevalence of cancer (non-skin) or melanoma among adults. The prevalence in Massachusetts is 8.3%.

Description: Cancer is a group of similar diseases in which some of the body's cells grow uncontrollably and spread to other parts of the body. There are many types of cancer that affect different parts of the body. At least two-thirds of cancer cases are caused by external factors. These factors include tobacco and alcohol use, diet, sun exposure, and certain infections. Exposure to certain chemical pollutants or radiation is also a risk factor. This indicator represents how many people are living with cancer.

Indicator type: *Sensitive Populations*

Rationale: Cancer is the leading cause of death in Massachusetts.[55] Cancer is a group of chronic diseases that can start in any organ or tissue of the human body. Cancers are also called malignant neoplasms or tumors. Cancer develops when abnormal cells grow uncontrollably and go beyond their usual boundaries to invade adjoining body parts or spread to other organs. Lung, colorectal, and breast cancers are common types of cancers. Cancer can happen in all age groups. Among 30-50% of all cancers can be prevented.[56] Multiple environmental risk factors have been associated with cancer development, such as air, soil, and water pollution, metals, pesticides, sun exposure, chemicals, and radon, among others.[57] In those patients diagnosed with cancer, environmental pollutants can worsen the prognosis.

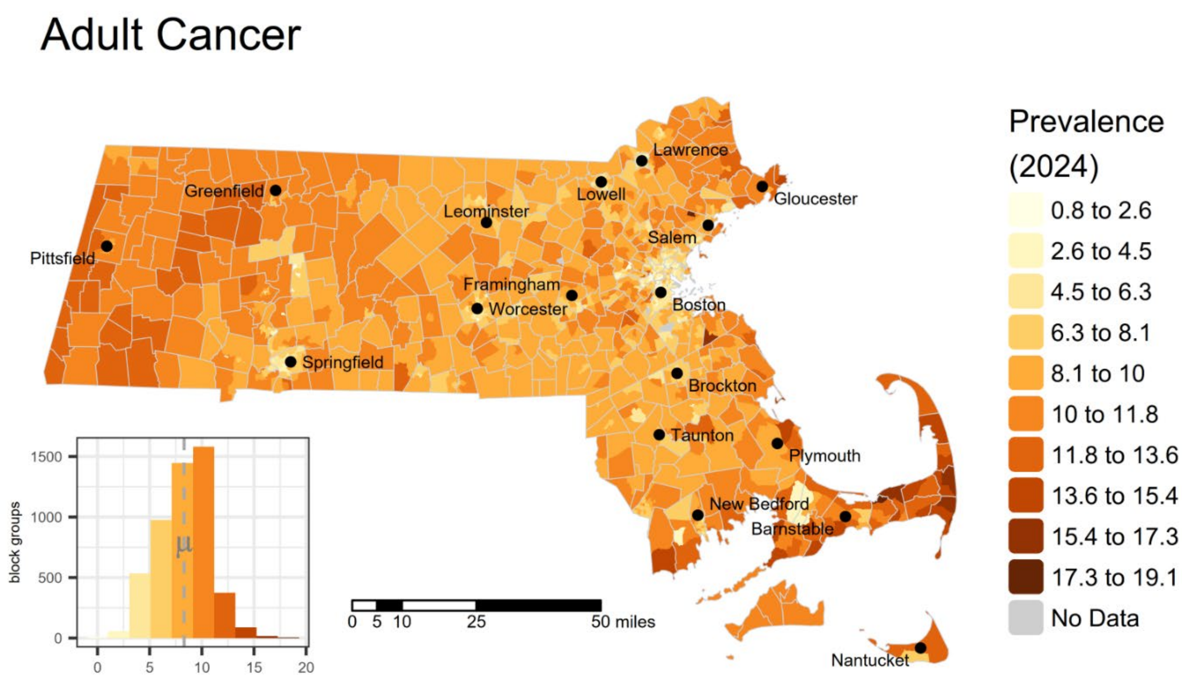
Method: Data provided at census tract level. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2024

Originating geographic scale of data: census tracts

Source: CDC PLACES Health Outcomes at https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data

Figure 31 - Map of Adult Cancer raw values



Population Characteristics - Socioeconomic Factors

Adults Without a High School Diploma

Definition: Percent of people age 25 or older whose education is less than a high school diploma.

Description: How much education a person has is associated with wealth, health, length of life, and exposure to pollution. People with less education are more likely to be affected by pollution because of where they live or work.

Indicator type: *Socioeconomic Factors*

Rationale: Educational attainment is an important independent predictor of health.[58] Individuals with lower education in the US have a lower life expectancy,[59] are more likely to be obese,[60] and are more likely to experience psychiatric disorders compared to individuals with higher education.[61] Education is often inversely related to the degree of exposure to indoor and outdoor pollution. Several studies have associated educational attainment with susceptibility to the health impacts of environmental pollutants. For example, individuals without a high school education appear to be at higher risk of mortality associated with particulate air pollution than those with a high school education.[62] There is also evidence that the effects of air and traffic-related pollution on respiratory illness, including childhood asthma, are more severe in communities with lower levels of education.[63] In studies evaluating air pollution related risks of adverse birth outcomes, mothers with low educational attainment were found to be more vulnerable.[64] While there is a positive association between educational attainment and health, racial and ethnic minorities gain fewer health benefits from educational attainment than Whites.[65]

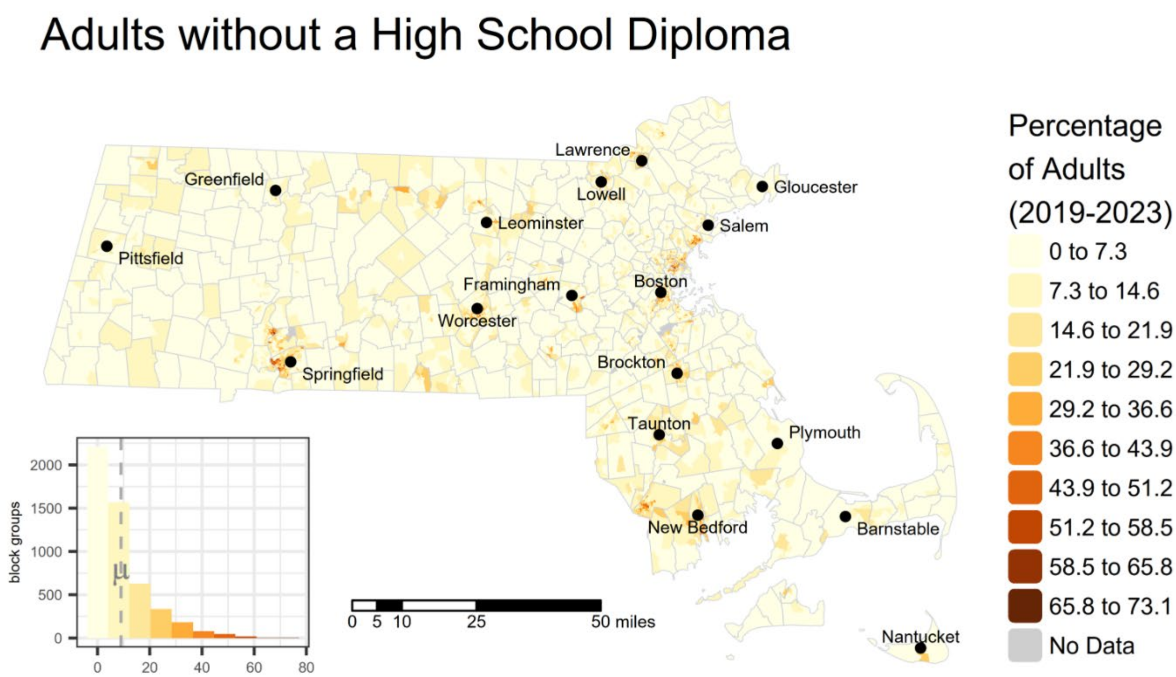
Method: The ACS education information is captured in the table Educational Attainment for the Population 25 Years and Over (ACS Table ID: B15003) at the census block group level. Percentage with less than a high school education is calculated as the sum of adults with 12th grade education but no diploma or lower level of education, divided by the total number of adults age 25 and older.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Source: US American Community Survey 5-year Estimates for 2019 – 2023.

Figure 32 - Map of Adults without a High School Diploma raw values



Housing Burdened Low Income Households

Definition: Percent of households that are both low income (making less than 80% of the HUD Area Median Family Income) and severely burdened by housing costs (paying greater than 50% of their income to housing costs).

Description: Areas where low-income households may be stressed by high housing costs can be identified through the Housing and Urban Development (HUD) Comprehensive Housing Affordability Strategy (CHAS) data. This indicator measures households earning less than 80% of HUD Area Median Family Income by county and paying greater than 50% of their income to housing costs. The indicator takes into account the regional cost of living for both homeowners and renters, and incorporates the cost of utilities. CHAS data are calculated from US Census Bureau's American Community Survey (ACS).

Indicator type: *Socioeconomic Factors*

Rationale: Housing cost-burdened communities are impacted by a reduced economic capacity to satisfy basic needs. Economic capacity also impacts environmental exposures, household quality, and environmental health resilience (e.g., health status, lifestyle, access to health services).[66] Housing cost-burdened communities tend to live in proximity to areas with higher environmental hazards, such as industrial sites, waste management facilities, and high-traffic density areas. Environmental inequities in housing cost-burdened communities are important, especially when other social factors converge (e.g., communities of color, lower levels of education).

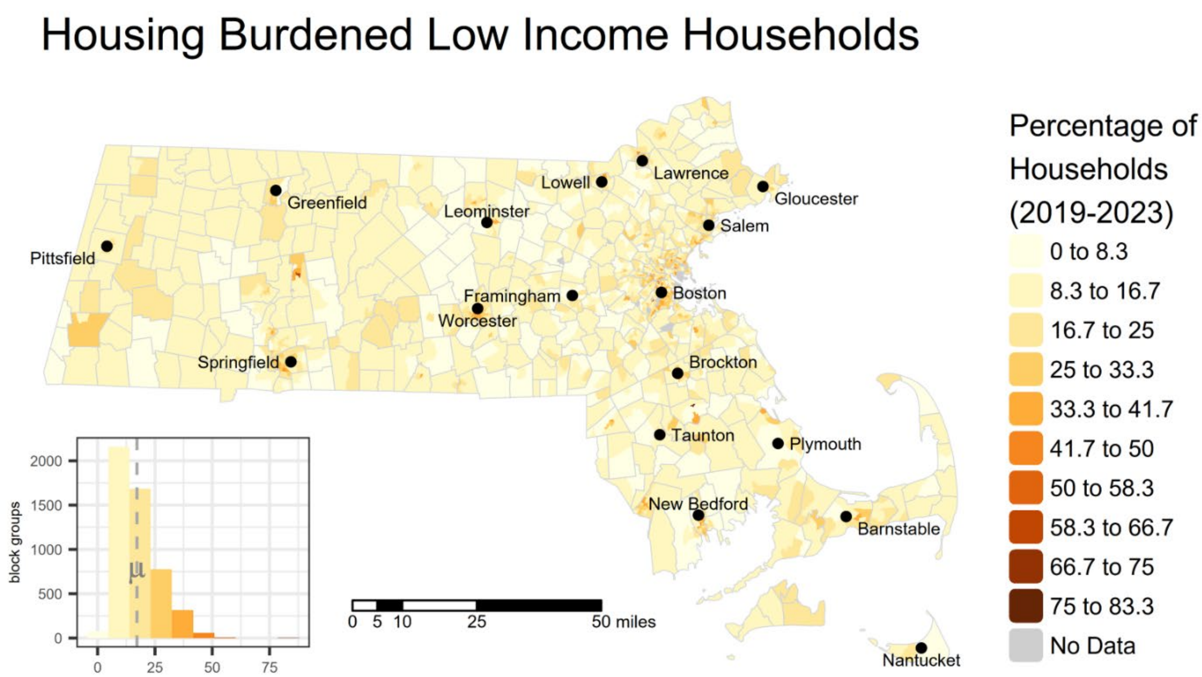
Method: Data provided at census tract level. All census block groups received the percentage value for the census tract in which they are located.

Data years: 2019 – 2023

Originating geographic scale of data: census tract

Source: US Department of Housing and Urban Development CHAS (Comprehensive Housing Affordability Strategy) at https://www.huduser.gov/portal/datasets/cp.html

Figure 33 - Map of Housing Burdened Low Income Households raw values



Linguistic Isolation

Definition: Percentage of limited English-speaking households.

Description: A limited English-speaking household is defined as a household in which no one age 14 or over speaks English at least "very well" as reported in the U.S. Census Bureau's ACS. Limited English proficiency affects how well people can access services and information in their community, such as public notices about environmental contamination. Linguistic isolation describes individuals and households that have limited English proficiency or speak languages other than English at home.

Indicator type: *Socioeconomic Factors*

Rationale: Linguistic isolation indicates communities that could be less resilient to climate and environmental risks.[67] Those communities might be less informed about environmental risks (e.g., air quality alerts or water violations) and climate risks (e.g., heat, flood, drought, and wildfire alerts) when information about those risks is only made available in English. Furthermore, linguistically isolated households may also have less access to health information (e.g., healthy lifestyles or disease screening), health care services (e.g., insurance or disease treatment), and environmental health actions (e.g., incentives for housing weatherization or pollution clean-ups). Linguistic isolation is highly correlated with income, and as mentioned in the low-income indicator, income impacts environmental exposures, household quality, and environmental health resilience.

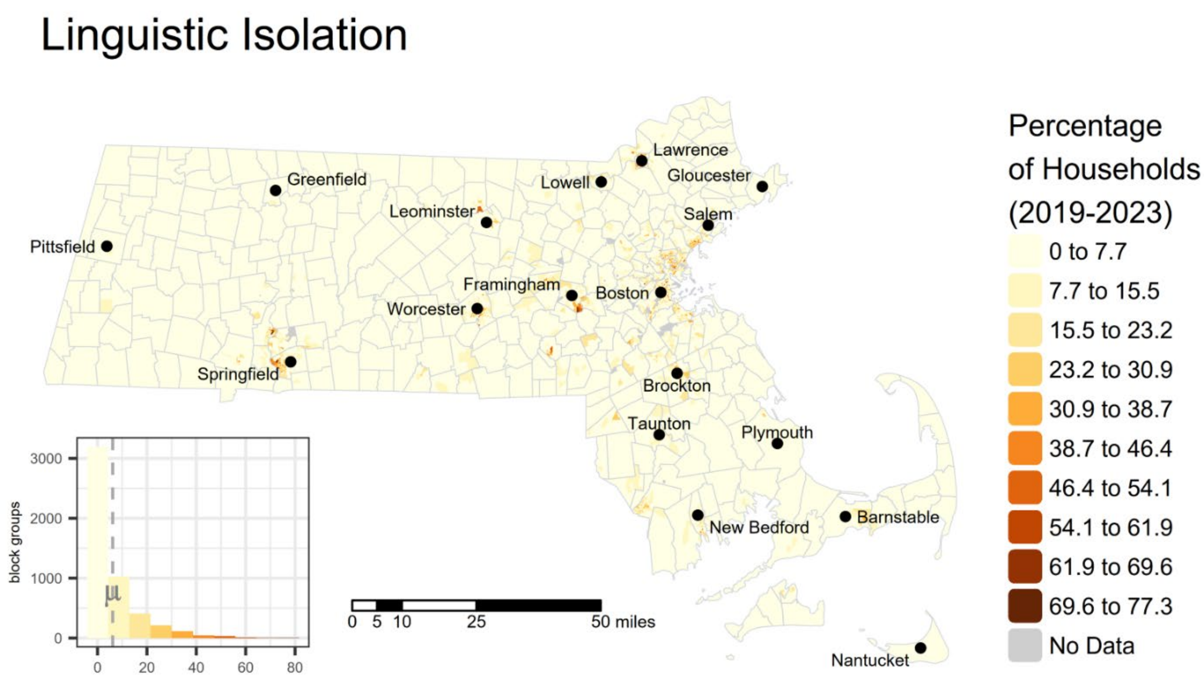
Method: The ACS limited English speaking household information is captured in the table Household Language by Household Limited English Speaking Status (ACS Table ID: C16002) at the census block group level. Percentage limited English speaking households is computed as the sum of Limited English speaking households whose primary language spoken at home is Spanish, Other Indo-European languages, Asian/Pacific Island languages, and other languages, divided by the total number of households.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Source: US American Community Survey 5-year Estimates for 2019 – 2023

Figure 34 - Map of Linguistic Isolation raw values



Median Household Income

Definition: Median household income in the past 12 months (in 2023 inflation-adjusted dollars).

Description: Median household income describes the household income within a census block group that is the median, or 50th percentile, for that block group. In other words, 50% of households within a given census block group have incomes below the median value, and 50% of households are above the median value for that census block group.

Indicator type: *Socioeconomic Factors*

Rationale: Household income is an important social determinant of health. Numerous studies have suggested that lower income populations are more likely than wealthier populations to experience adverse health outcomes when exposed to environmental pollution or other stressors. Median household income, rather than mean or average household income, is unaffected by outliers or extreme low or high household income values within a given block group.

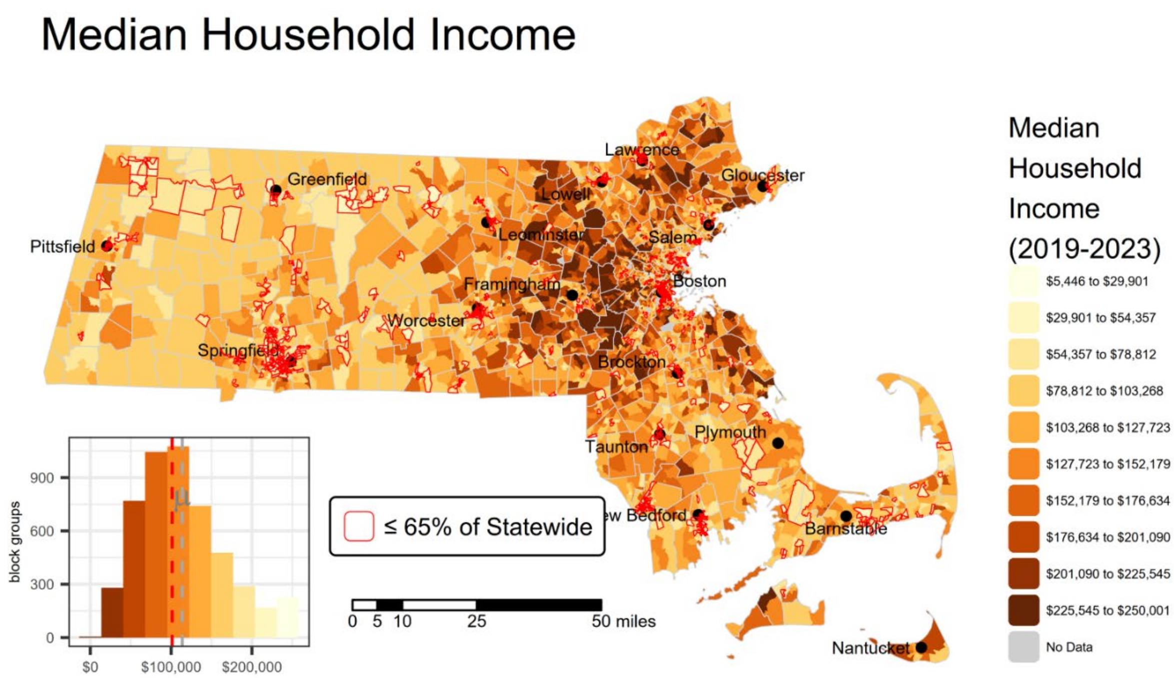
Method: The ACS median household income information is captured in the table Median Household Income (ACS Table ID: B19013) at the census block group level. 439 census block groups are missing median household income values, likely due to data suppression as a result of higher uncertainty or low response rates. 400 of these values were imputed using the median household income of the census tract in which they reside for the same time period.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Source: US American Community Survey 5-year Estimates for 2019 – 2023

Figure 35 - Map of Median Household Income raw values



Poverty

Definition: Percent of households whose income is less than or equal to twice the federal poverty level.

Description: Income is strongly associated with health outcomes, environmental exposures, and access to, or quality of services. This indicator describes how many people in a community live at or below twice the federal poverty level. The poverty level is a national number and the same across all geographic regions. To accommodate differences in the varying costs of living across the United States and other factors, analysts typically use twice the poverty level to capture low-income households, especially in high-cost areas such as Massachusetts. In 2023, twice the federal poverty threshold meant an annual household income of $62,400 or less for a family of four.

Indicator type: *Socioeconomic Factors*

Rationale: Income correlates with access to better services and goods and is associated with better health outcomes and life expectancy. Income is also associated with housing quality and climate resilience. Low-income communities tend to live in areas with more environmental pollutants (e.g., industrial sites, waste management facilities, high-traffic areas) and climate hazards (heat islands, floods, droughts, and wildfires). Moreover, low income communities also suffer from higher disease rates, injuries, and premature mortality and have less access to public health and health care opportunities.[68]

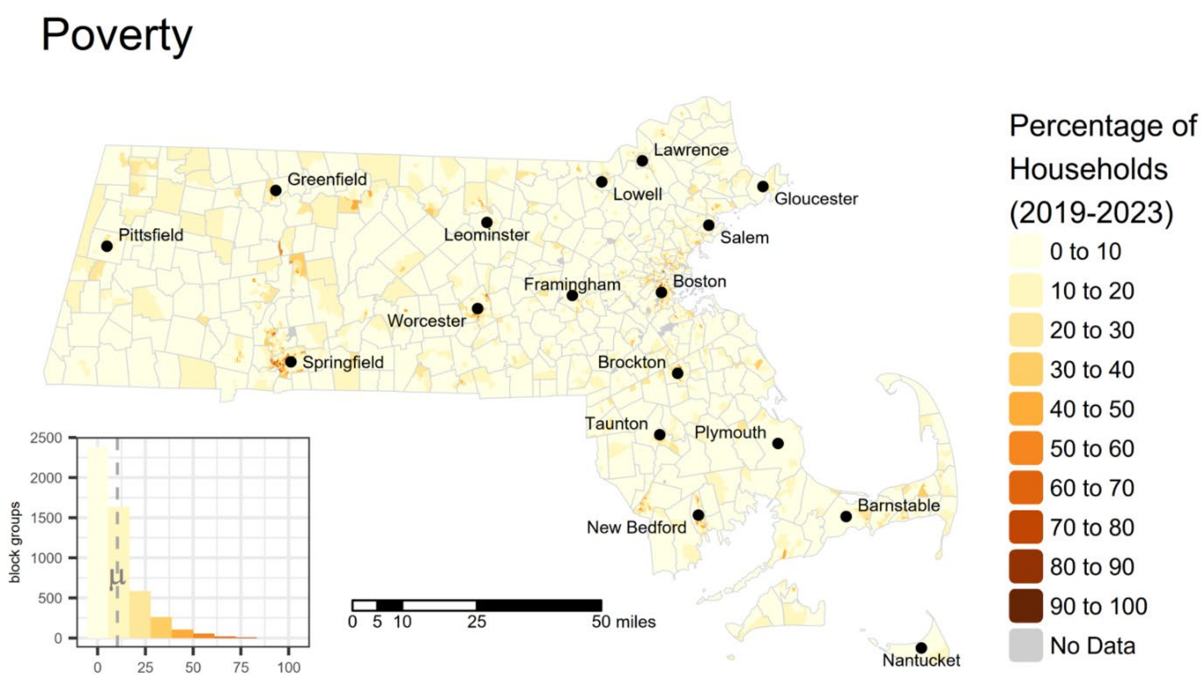
Method: The ACS low income information is captured in the table Ratio of Income to Poverty Level in the Past 12 Months (ACS Table ID: C17002) by census block group.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Source: US American Community Survey 5-year Estimates for 2019 – 2023.

Figure 36 - Map of Poverty raw values



Unemployment

Definition: Percentage of the civilian population over the age of 16 who are unemployed and eligible for the labor force. Excludes retirees, students, homemakers, institutionalized persons except prisoners, those not looking for work, and military personnel on active duty.

Description: Unemployment impacts people's health, well-being, and quality-of-life. Many people live in poverty because they cannot find employment. In addition, some workforce participants are "underemployed", including involuntary part-time employment, poverty-wage employment, and/or insecure employment. In either case, these people struggle to afford healthy foods, health care, and safe, affordable housing, and lack the time and resources for a healthy lifestyle (e.g., exercise, mediation, stress reduction).

Indicator type: *Socioeconomic Factors*

Rationale: Unemployment has a wide range of effects on health which contribute to the burden placed on vulnerable communities. It has been shown to negatively impact mental and physical health. Higher rates of unemployment are associated with overall mortality, as well as mortality specifically due to transport accidents, poisonings (which include drug overdoses), and suicides.[69] Unemployment is also associated with increases in physical morbidity as well as mortality.

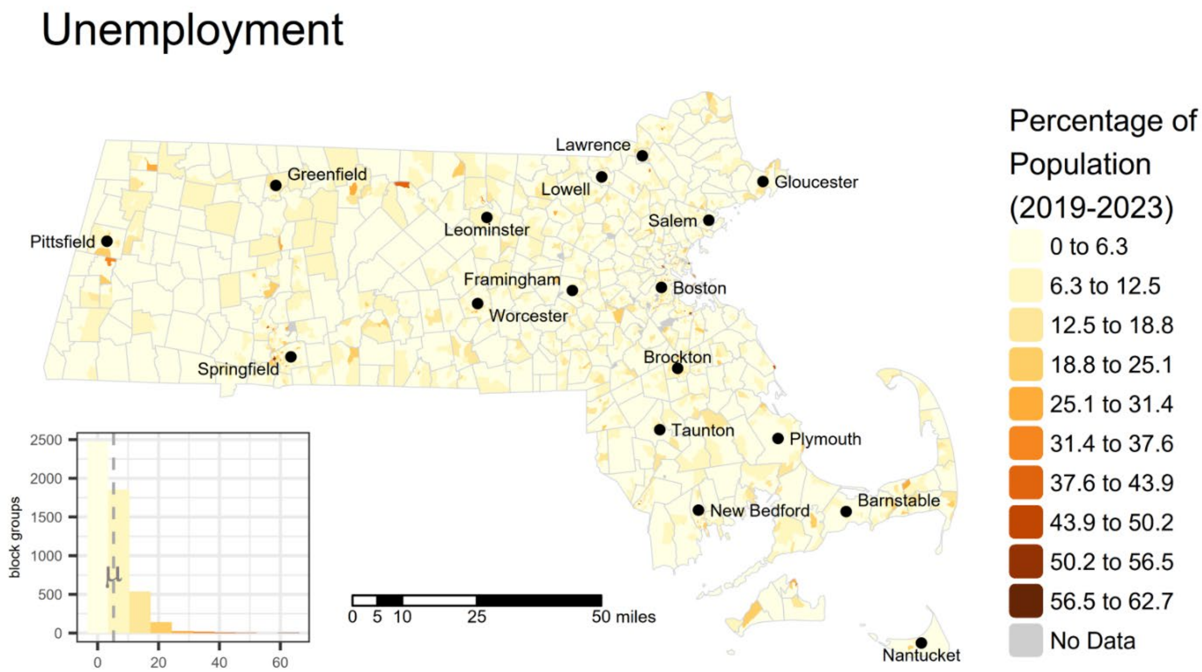
Method: The ACS unemployment information is captured in the table Employment Status for the Population 16 Years and Over (Table ID: B23025).

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Source: US American Community Survey 5-year Estimates for 2019 – 2023.

Figure 37 - Map of Unemployment raw values



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